

DEEP LEARNING BASED MULTI-CLASS SKIN LESION DIAGNOSIS USING DERMOSCOPIC IMAGES

A PROJECT REPORT

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INTERNAL EXAMINER**EXTERNAL EXAMINER**

DECLARATION

We affirm that the project report entitled “**DEEP LEARNING BASED MULTI-CLASS SKIN LESION DIAGNOSIS USING DERMOSCOPIC IMAGES**” being submitted in partial fulfilment of the requirements for the award of Master of Computer Applications is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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ABSTRACT

Skin cancer is one of the leading causes of mortality worldwide, with its incidence increasing significantly each year. Early and accurate detection of malignant skin lesions is crucial for effective treatment and improved patient survival. Dermoscopic imaging is widely used as a non-invasive diagnostic tool for identifying various skin conditions, including melanoma and other cancerous lesions. However, manual analysis of dermoscopic images is often time-consuming, subjective, and dependent on the expertise of dermatologists, which may lead to variability in diagnosis, especially in large-scale or complex cases. To address these challenges, this research presents a deep learning-based framework for reliable multi-class skin lesion classification.

The proposed framework evaluates multiple advanced deep learning architectures, including EfficientNetV2-S, ConvNeXt, and DenseNet, along with a hybrid model that combines EfficientNetV2-S and ConvNeXt. EfficientNetV2-S provides optimized performance with improved training efficiency and accuracy, while ConvNeXt leverages modern convolutional design principles for enhanced feature representation. DenseNet facilitates better feature propagation through dense connectivity, improving learning efficiency. The proposed hybrid model integrates the strengths of EfficientNetV2-S and ConvNeXt to capture diverse and complementary feature representations, thereby improving classification robustness and consistency.

The models are trained and evaluated on the HAM10000 dermoscopic image dataset, focusing on four clinically important classes: melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec). Standard preprocessing techniques, including image resizing, normalization, and data augmentation, are applied to improve dataset quality and generalization. Experimental results show that EfficientNetV2-S achieves 98% accuracy, ConvNeXt achieves 95%, and DenseNet achieves 90%, while the hybrid model achieves 94% with more balanced performance across all classes. Furthermore, high Area Under the ROC Curve (AUC) values indicate strong discriminative capability. Gradient-weighted Class Activation Mapping (Grad-CAM) is utilized to highlight important regions influencing model predictions, improving interpretability.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
CNN	Convolutional Neural Network
AI	Artificial Intelligence
DL	Deep Learning
AUC	Area Under Curve
ROC	Receiver Operating Characteristic
TPR	True Positive Rate
FPR	False Positive Rate
ReLU	Rectified Linear Unit
NV	Melanocytic Nevi
MEL	Melanoma
BCC	Basal Cell Carcinoma
AKIEC	Actinic Keratosis / Intraepithelial Carcinoma
DF	Dermatofibroma
CAM	Class Activation Mapping
Grad-CAM	Gradient-weighted Class Activation Mapping
EfficientNetV2-S	Efficient Network Version 2 – Small
ConvNeXt	Convolutional Neural Network – Next Generation
DenseNet	Densely Connected Convolutional Network

F1-Score	Harmonic Mean of Precision and Recall
RGB	Red Green Blue
PX / Pixels	Picture Elements
224×224	Image Resolution (Height × Width)
LR	Learning Rate
ADAM	Adaptive Moment Estimation (Optimizer)
Epochs	Number of Training Iterations
JPG / JPEG	Joint Photographic Experts Group
GPU	Graphics Processing Unit
Collab	Google Colaboratory

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Skin cancer continues to be one of the most common forms of cancer globally, presenting a serious public health concern. It originates from the uncontrolled growth of abnormal skin cells, which is primarily triggered by prolonged exposure to ultraviolet (UV) radiation, along with genetic predisposition and various environmental factors. Among the most frequently diagnosed types are melanoma, basal cell carcinoma and benign conditions such as melanocytic nevi and actinic keratosis. Of these, melanoma is particularly dangerous due to its aggressive nature and high potential for metastasis, making early diagnosis critical to improving patient survival.

Dermoscopy is a non-invasive imaging technique widely used in dermatology that allows clinicians to examine subsurface skin structures with far greater clarity than standard visual inspection. It reveals important diagnostic features such as pigment networks, streaks, globules and vascular patterns that are otherwise invisible to the naked eye. Despite its advantages, interpreting dermoscopic images demands significant expertise and even experienced dermatologists may produce inconsistent results when evaluating visually similar lesions.

In recent years, artificial intelligence and deep learning have become powerful tools for automating pattern recognition tasks, including medical image analysis. Convolutional Neural Networks (CNNs) are highly effective at learning important feature representations directly from raw image data, reducing the need for manual feature engineering. These capabilities make deep learning models well-suited for dermoscopic image analysis, where subtle differences in texture, color and lesion boundaries play a critical role in accurate diagnosis.

To offer a contribution towards this topic, this study aims at creating a deep learning-based system to categorise multiple skin lesion classifications based on dermoscopic images in the HAM10000 dataset. More exactly, this research focuses on four important skin lesions: melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc) and actinic keratosis (akiec). For feature extraction and image classification, the approach leans on two strong models EfficientNetV2 and ConvNeXt. And to make sure the predictions actually make sense

to the people using them, Grad-CAM is used to visually point out which parts of the image the model paid attention to when reaching its decision.

1.2 PROBLEM STATEMENT

Skin cancer is one of the most rapidly growing health concerns worldwide, with millions of new cases diagnosed each year across all age groups and demographics. Among the various forms of skin cancer, melanoma stands out as the most dangerous due to its aggressive metastatic behavior, while conditions such as Basal Cell Carcinoma and Actinic Keratosis, though less immediately fatal, carry significant morbidity if left undetected. The primary diagnostic tool for evaluating suspicious skin lesions is dermoscopy, a non-invasive imaging technique that reveals subsurface skin structures invisible to the naked eye. However, accurate interpretation of dermoscopic images demands years of specialized clinical training. In many parts of the world, particularly in rural and resource-limited healthcare settings, qualified dermatologists are scarce or entirely unavailable, leaving large patient populations without access to timely and reliable skin lesion evaluation. This gap between diagnostic need and specialist availability creates a critical public health challenge that urgently calls for automated, scalable solutions.

The complexity of the diagnosis is further compounded by the striking visual similarity that exists between different lesion types. Conditions such as melanoma and dysplastic nevi, or actinic keratosis and squamous cell carcinoma, can appear nearly indistinguishable even under dermoscopic examination, leading to inter-observer disagreement among experienced clinicians. Subtle differences in pigmentation patterns, border irregularity, color heterogeneity and lesion texture must be carefully weighed to arrive at an accurate classification. This inherent ambiguity means that even specialist dermatologists are susceptible to diagnostic errors, particularly under time pressure or when managing high patient volumes. A missed melanoma diagnosis, for instance, can have life-threatening consequences, while unnecessary biopsies driven by false positives impose financial burdens and patient anxiety. The stakes of misclassification in this domain are therefore exceptionally high, reinforcing the need for a more consistent and objective diagnostic approach.

While significant progress has been made in the field of computer-aided diagnosis for skin lesions, existing automated systems continue to fall short of clinical expectations in several important ways. A large proportion of published models address only binary classification tasks, distinguishing broadly between malignant and benign lesions without differentiating

between specific lesion subtypes. This coarse categorization fails to reflect the nuanced reality of clinical practice, where a physician must not only determine whether a lesion is dangerous but also identify its exact type to prescribe the appropriate treatment. Furthermore, many models are trained on datasets that are heavily imbalanced, with some lesion categories vastly outnumbering others. When these imbalances are not properly addressed during training, models tend to develop a bias toward majority classes while performing poorly on the minority classes that are often the most clinically critical. Poor generalization to unseen patient data, sensitivity to image acquisition conditions, and limited robustness across different demographic groups further restrict the real-world applicability of many existing systems.

A particularly serious limitation that cuts across nearly all current automated systems is the lack of interpretability. Deep learning models, despite their strong predictive performance, are often described as black boxes whose internal decision-making logic is opaque to end users. In a clinical setting, a physician cannot reasonably act on a model's output without some understanding of why a particular prediction was made and which features of the image drove that decision. This absence of transparency significantly erodes clinician trust and remains one of the primary barriers to the adoption of AI-based diagnostic tools in real healthcare environments. There is therefore a clear and pressing need for a comprehensive deep learning framework that goes beyond binary classification, handles multiple lesion types reflective of actual clinical practice, addresses class imbalance effectively, generalizes well across diverse patient populations, and integrates explainability mechanisms that make model predictions transparent, trustworthy, and actionable for clinical decision-making.

1.3 OBJECTIVES OF THE RESEARCH

This project aims to accomplish the following objectives:

- To develop a deep learning-based framework for the automated multi-class classification of skin lesions from dermoscopic images.
- To utilize the HAM10000 dataset and classify skin lesions into four clinically relevant categories: melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec).
- To implement and evaluate three deep learning architectures EfficientNetV2-S, ConvNeXt, and DenseNet for their classification performance.
- To propose a hybrid deep learning model that combines EfficientNetV2 and ConvNeXt to achieve more balanced and robust multi-class classification.

- To apply preprocessing techniques and data augmentation to improve model generalization and reduce the impact of class imbalance.
- To incorporate Grad-CAM visualization for model interpretability, enabling transparent and clinically meaningful predictions.

1.4 SCOPE AND LIMITATIONS

SCOPE

- **Multi-Class Skin Lesion Classification:** This study focuses on classifying dermoscopic images into four clinically significant categories: Melanocytic Nevi, Melanoma, Basal Cell Carcinoma, and Actinic Keratosis. The multi-class approach reflects real clinical diagnostic scenarios more accurately than binary classification systems.
- **HAM10000 Dataset Utilization:** The project employs the publicly available HAM10000 dataset, which comprises 10,015 high-resolution dermoscopic images collected from real clinical environments. This dataset provides a diverse and medically validated foundation for training and evaluating the proposed models.
- **Deep Learning Architecture Evaluation:** Multiple state-of-the-art deep learning architectures, including EfficientNetV2-S, ConvNeXt Tiny, and DenseNet, are individually implemented and evaluated. This comparative analysis enables a systematic understanding of each model's strengths and limitations for skin lesion classification.
- **Hybrid Dual-Branch Model Design:** A novel hybrid architecture combining EfficientNetV2 and ConvNeXt is proposed to leverage complementary feature representations from both networks. The dual-branch design with feature-level fusion aims to improve classification consistency across all lesion categories.
- **Preprocessing and Data Augmentation Pipeline:** A structured preprocessing pipeline including resizing, normalization, and ImageNet-standard standardization is applied to all input images. Data augmentation techniques such as random flipping, rotation, zooming, and color jitter are incorporated to improve model generalization and reduce overfitting.
- **Class Imbalance Management:** The study addresses the significant class imbalance present in the HAM10000 dataset, where Melanocytic Nevi substantially outnumbers

other lesion types. Augmentation strategies and balanced evaluation metrics are employed to ensure fair and reliable performance assessment across all classes.

- **Grad-CAM Explainability Integration:** Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated into the classification pipeline to generate visual heatmaps of model predictions. These heatmaps highlight the lesion regions that most influence the model's decisions, making the system interpretable and suitable for clinical trust-building.
- **Performance Evaluation and Benchmarking:** All models are rigorously evaluated using accuracy, precision, recall, F1-score, confusion matrix analysis, and ROC-AUC curves. The evaluation framework ensures a comprehensive and clinically meaningful comparison of model performance across all four lesion categories.

LIMITATIONS

- The study focuses only on four selected skin lesion categories from the HAM10000 dataset — melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec). Although these classes are clinically important, other lesion types available in the complete HAM10000 dataset are not included in the current research, which limits the overall classification scope of the system.
- The proposed deep learning models are trained and evaluated using a specific dermoscopic image dataset under controlled conditions. As a result, the trained models may not always generalize effectively to images collected from different hospitals, imaging devices, skin tones, or real-world clinical environments without additional retraining or fine-tuning.
- Clinical metadata such as patient age, gender, lesion location, and medical history are not incorporated into the classification framework. Since dermatologists often use both visual examination and patient-related clinical information for diagnosis, the absence of metadata may limit the overall diagnostic capability of the proposed system.
- The proposed framework is designed as a computer-aided decision-support system to assist dermatologists in skin lesion analysis. It is not intended to replace professional medical diagnosis, clinical expertise, or histopathological examination, and final clinical decisions should always be made by qualified healthcare professionals.
- The HAM10000 dataset contains significant class imbalance, where some lesion categories have far more samples than others. Although preprocessing and data

augmentation techniques are applied to reduce this imbalance, minority classes may still receive fewer representative samples during training, which can affect classification performance and model reliability for underrepresented lesions.

1.5 SIGNIFICANCE OF THE STUDY

This study makes several meaningful contributions to the field of automated skin lesion analysis. By combining two state-of-the-art CNN architectures into a hybrid framework, the project demonstrates that feature diversity and model complementarity lead to more balanced and reliable classification than any single architecture can achieve alone. The integration of Grad-CAM visualization adds an important layer of interpretability, producing results that are not only accurate but also explainable a quality that is essential for acceptance in clinical settings.

From a practical standpoint, the developed system has the potential to assist dermatologists in clinical workflows, particularly in high-volume settings where timely and consistent diagnosis is critical. It can serve as a screening tool that flags potentially malignant lesions for further review, thereby reducing diagnostic delays and improving patient care. The project also contributes to the growing body of research that validates deep learning as a viable and powerful approach to medical image analysis, particularly in the domain of dermatological diagnosis.

SUMMARY

This chapter presented the background, objectives, scope, limitations, and significance of the proposed skin lesion classification project. Skin cancer is a major global health issue, where early and accurate detection plays an important role in effective treatment. Traditional diagnostic methods often depend on specialist expertise and may lead to subjective interpretation. To address these challenges, deep learning techniques are applied for automated dermoscopic image analysis. The project uses the HAM10000 dataset containing four clinically important lesion categories: nv, mel, bcc, and akiec. Advanced architectures such as EfficientNetV2, ConvNeXt, and DenseNet are implemented and analyzed for classification performance. A hybrid model combining EfficientNetV2 and ConvNeXt is also proposed to improve feature extraction and prediction accuracy. In addition, Grad-CAM visualization is incorporated to enhance model interpretability by highlighting important lesion regions.

CHAPTER 2

LITERATURE REVIEW

2.1 REVIEW OF PREVIOUS WORK

Over the past decade, significant research effort has been directed toward automating skin lesion diagnosis using deep learning. A wide range of architectures, techniques, and datasets have been explored, each contributing valuable insights to the field.

Ahmad et al. [1] (2023) Ahmad et al. proposed a multiclass skin lesion recognition framework combining deep learning with explainable AI. The study used the HAM10000 and ISIC2018 datasets, fusing deep features extracted from Xception and ShuffleNet. A Butterfly Optimization Algorithm was then applied for feature selection to reduce redundancy. The approach achieved 99.3% accuracy on HAM10000 and 91.5% on ISIC2018, demonstrating strong benchmark performance. It also improved interpretability by making predictions more transparent to users. However, using two backbone networks significantly increased computational cost and training complexity. The model also relied heavily on benchmark datasets without broader clinical validation.

Wang et al. [2] (2023) Wang et al. proposed a double-branch network designed for clinical skin lesion classification across multiple disease categories. The model used the CSLI dataset, built from PAD-UFES-20 and newly collected clinical images, totaling 3,361 samples across six categories. The dual-branch design allowed the network to learn both shallow and deep feature representations simultaneously. The best reported result for the proposed DBN reached 91.58% accuracy, showing that the architecture improves over single-branch alternatives. Its strength lies in being more suited to real clinical images rather than dermoscopic-only datasets. However, the dataset remains relatively small and the model struggles with similar-looking lesion categories and class imbalance. Illumination and image quality variation also affected the model's sensitivity.

El-Khatib et al. [3] (2023) El-Khatib et al. presented a multi-network decision fusion system for melanoma detection. The model was trained on ISIC 2020 and DermIS datasets, combining several CNNs including DarkNet-53, DenseNet-201, Inception-V3, ResNet-50, and ResNet-101, among others. An F1-score-based fusion strategy was used to integrate predictions and support integration with electronic health records. The ensemble approach helped reduce the individual weaknesses of each architecture by leveraging their collective

strengths. Its practical contribution is improved reliability for melanoma screening within clinical decision support systems. However, the system required high computational resources and longer inference time due to the large model ensemble. Such complexity may limit its use in real-time or resource-constrained clinical environments.

Olayah et al. [4] (2023) Olayah et al. studied dermoscopy image analysis using a combined CNN feature extraction approach. The method used the ISIC 2019 dataset and extracted features from lesion regions before applying a machine learning classifier. The hybrid CNN-feature approach achieved 92.20% accuracy, outperforming simpler single-model pipelines on the same dataset. The study demonstrated that combining multiple deep CNN representations can improve early lesion detection performance. Its practical contribution lies in showing how fused feature strategies add value in dermoscopic image analysis. However, the method still relied partly on manual or semi-manual processing steps, which reduced full automation potential. It also faced difficulty managing class imbalance across multiple lesion categories.

Kim et al. [5] (2023) Kim et al. presented a hybrid CNN that integrated edge, color and texture features for skin lesion classification. The study used the ISIC2017 and ISIC2018 datasets, combining handcrafted lesion cues with deep convolutional learning. On ISIC2018, the proposed method achieved 94.9% accuracy, while it reached 97.1% on the melanoma and seborrheic keratosis task using ISIC2017. The practical contribution is showing how structured feature extraction and CNN learning can be combined to capture diagnostically relevant patterns. This multi-cue approach helped the model learn more meaningful representations than single-feature CNN systems. However, the feature engineering process may discard some information and add extra preprocessing complexity to the pipeline. Such preprocessing steps can also introduce inconsistencies when applied to varied clinical image sources.

Manole et al. [6] (2024) Manole et al. explored transfer learning for dermatological diagnosis using EfficientNet-B3 as the backbone model. The dataset included 8,222 skin images drawn from the authors' collection and the ISIC2019 archive. The model achieved 95.4% validation accuracy in a four-class setting, demonstrating strong adaptability with limited annotated data. However, performance dropped to 88.8% when the task was extended to six lesion categories, highlighting scalability issues. This shows that a single architecture may not be sufficient for more complex multi-class diagnostic problems. The study is practically valuable for demonstrating how pretrained models can be fine-tuned under data-

scarce conditions. Still, broader validation and architecture adaptation would be needed for clinical deployment.

Gamil et al. [7] (2024) Gamil et al. proposed an AdaBoost-based hybrid framework for skin cancer detection and classification. The method used EfficientNet-B0 as a deep feature extractor and applied AdaBoost for the final decision stage, evaluated on the ISIC dataset. The hybrid approach achieved 88.4% accuracy, outperforming standalone CNN and SVM baselines reported in the same paper. Its practical contribution is that classical boosting strategies can still enhance deep feature-based skin cancer diagnosis. The combination of deep learning and ensemble methods provided improved classification reliability over single-model systems. However, the model depended heavily on parameter tuning and was primarily explored in a binary classification setting. This limits its direct generalizability to multi-class skin lesion diagnosis tasks.

Azeem et al. [8] (2024) Azeem et al. developed SkinLesNet, a multilayer CNN designed for skin lesion and melanoma classification. The study used PAD-UFES-20-Modified, HAM10000, and ISIC2017 datasets, focusing on three common lesion types. On the PAD-UFES-20-Modified dataset, the model achieved 96% accuracy, demonstrating strong performance on smartphone-captured clinical images. Its practical contribution is that the lightweight network design makes it suitable for accessible and mobile skin screening applications. The study reflects the growing need for compact diagnostic tools that go beyond hospital-grade dermoscopic setups. However, the evaluation covered only a limited number of lesion classes and remained sensitive to dataset imbalance. Further extension to a broader class set and more diverse datasets is needed.

Di et al. [9] (2024) Di et al. proposed ECRNet, a hybrid network for skin cancer identification that combined convolutional and transformer-based learning. Evaluated on the ISIC2018 dataset, the architecture was designed to capture both local texture details and global contextual relationships in skin images. The model achieved 92.12% accuracy, demonstrating that the hybrid design offers advantages over standard CNN-only systems. Its practical contribution is a well-balanced architecture that addresses the complementary strengths of CNNs and transformers. The model is more flexible and powerful for handling complex lesion patterns compared to simpler single-path designs. However, the hybrid design increased model size and inference cost, which may affect deployment in resource-limited settings. Further optimization would be needed for efficient real-time clinical use.

Almufareh et al. [10] (2024) Almufareh et al. focused on melanoma identification using a fine-tuned convolutional neural network with transfer learning. The study used dermoscopic datasets including DermIS, DermQuest, and ISIC2019 to evaluate classification performance across different data sources. Transfer learning improved accuracy by approximately 5% on DermIS, 6% on DermQuest, and 0.81% on ISIC2019 over earlier baselines. The practical value of this work is that it supports early melanoma screening in realistic clinical and smartphone-based settings. The study confirmed that fine-tuning pretrained models is highly effective even when annotated medical data is limited. However, the dataset remained relatively small, and the work was limited to a few lesion categories. Broader clinical validation across more diverse lesion types is needed before practical deployment.

Kandhro et al. [11] (2024) Kandhro et al. evaluated an enhanced VGG19-based model called E-VGG19 for real-time skin cancer detection and classification. The study used the ISIC2019 dataset and combined VGG19 with traditional machine learning classifiers to improve feature extraction. The paper reported 88% training accuracy and 91% validation accuracy, showing that the enhanced backbone can support practical classification tasks. Its practical contribution is demonstrating that classic CNN architectures can still be effective for real-time screening when properly enhanced. The method maintained computational practicality while improving upon the standard VGG19 baseline. However, the model still depended on good-quality dermoscopic input and did not fully address generalization across all lesion classes. It also lacked stronger interpretability components compared to more recent explainable AI methods.

Gomathi and Arunachalam [12] (2024) Gomathi and Arunachalam introduced a skin lesion prediction and classification model based on modified LSTM combined with hybrid optimization strategies. The study used the ISIC challenge dataset containing 2,594 images and covered seven lesion categories in the classification task. The model achieved an impressive 99.31% accuracy and 98.25% recall, reflecting strong performance across the multi-class setting. Its practical contribution is a comprehensive pipeline that integrates preprocessing, segmentation, feature extraction, and optimized classification in one unified framework. The hybrid approach demonstrated that intelligent optimization methods can add value beyond conventional CNN-based pipelines. However, the method depends on several sequential processing stages, making it less direct than end-to-end deep learning approaches. It may also face challenges when scaling to large clinical image datasets.

Flosdorf et al. [13] (2024) Flosdorf et al. studied skin cancer detection using Vision Transformer models, comparing ViT variants with CNN baselines. The study used the HAM10000 dataset and analyzed attention maps to evaluate model interpretability alongside classification performance. The ViT-L16 variant achieved 92.79% accuracy and ViT-L32 reached 91.57%, confirming that transformer attention is effective for dermoscopic image classification. The practical contribution is demonstrating how transformers can capture broader contextual relationships in skin images that CNNs may miss. The study also showed that attention visualization can support clinical interpretability by highlighting relevant image regions. However, the work remained at the preprint stage with limited clinical validation, reducing confidence in its broader applicability. The melanoma recall also remained relatively modest, suggesting room for further improvement on minority classes.

Fiaz et al. [14] (2025) Fiaz et al. developed an explainable hybrid framework that performs lesion segmentation before multi-class classification. The method used HAM10000 dermoscopic images, with U-Net handling segmentation and EfficientNet-B0 performing classification. Grad-CAM was incorporated to improve the interpretability of predictions for clinical use. The model achieved a Dice score above 0.85 for segmentation and classification accuracy close to 85%. The joint use of segmentation, classification, and visual explanation in one workflow is its main practical strength. However, the system still faced challenges related to class imbalance, especially for melanoma cases. Overall performance also depended on accurate lesion localization during the segmentation stage.

Alotaibi et al. [15] (2025) Alotaibi et al. proposed a transfer learning approach enhanced with deep attention mechanisms for skin cancer detection. Built on the Xception architecture, the model was evaluated on the HAM10000 dataset in a binary classification setting. The attention component helped focus on the most informative lesion regions, reducing the influence of irrelevant background details. The model achieved 93.2% accuracy, 95.3% precision, 91.4% recall, and an F1-score of 93.3%, reflecting strong performance. Its main practical contribution is the ability to selectively emphasize discriminative features during classification. However, the model was not extended to a full multi-class diagnostic setting, limiting its direct clinical applicability. Further development would be needed to handle the full range of lesion types in dermatology.

Vieira et al. [16] (2025) Vieira et al. conducted a benchmarking study comparing deep learning approaches for automated skin lesion detection. The work evaluated standard CNN-style models and capsule-based architectures on dermoscopic skin lesion datasets, with the

best-performing method reaching 96.49% accuracy on HAM10000. The study helped researchers identify which architectures perform most effectively before building new models. Its practical value lies in providing a systematic comparison that guides future research directions in the field. The paper emphasized the growing role of pretrained CNNs and transfer learning in medical image analysis. However, the study did not introduce a new classification method of its own and thus mainly serves as a reference point. It remains useful for understanding the current state of the field rather than as a technical contribution.

Sohel et al. [17] (2025) Sohel et al. introduced SkinVisualNet, a hybrid deep learning model with integrated explainable AI components. While the paper focused on Lyme disease rash images rather than dermoscopic skin cancer classification, its methodology remains relevant to skin image analysis research. The model combined VGG19 and DenseNet201 and achieved 98.83% accuracy along with a 98.76% F1-score on the target task. Its strongest contribution is the integration of Grad-CAM and LIME to generate visual explanations that help build trust in clinical predictions. This shows how explainability tools can be embedded into hybrid deep learning pipelines for medical imaging. However, since the study is not a dermoscopic skin cancer classification task, direct comparisons with lesion-based studies are limited. The design principles are still transferable to broader skin image classification frameworks.

Thapar et al. [18] (2025) Thapar et al. proposed a hybrid FrCN-(U-Net) framework that combined lesion segmentation with classification for skin lesion analysis. The study used the ISBI-2017 dataset and linked an R-CNN classifier with U-Net segmentation to distinguish benign lesions from melanoma. The core idea was to isolate the lesion region first and then perform more focused and accurate classification. This pipeline approach was technically interesting as it integrated two complementary processing stages into a single workflow. However, the paper was later retracted, which significantly reduces its reliability as a reference for research comparison. The retraction means the reported results and methodology cannot be treated as stable or verified contributions. It should be mentioned only with caution and should not serve as a strong basis for further evaluation.

Zhang et al. [19] (2025) – DermViT Zhang et al. proposed DermViT, a diagnosis-guided Vision Transformer for skin lesion classification. The model used HAM10000 and ISIC-based dermoscopic images and incorporated clinical priors into the transformer attention process to improve robustness. It achieved 92.79% accuracy with ViT-L16 and 91.57% with ViT-L32, confirming that guided transformer methods perform strongly on dermoscopic tasks.

The practical contribution is that clinical guidance helps align the model's focus with medically relevant lesion features. This makes the system more robust and better aligned with real-world diagnostic patterns compared to standard ViT models. However, the architecture is heavier than most CNN alternatives and may be less suitable for lightweight or resource-constrained deployment. Further validation across diverse and minority lesion classes is also needed.

Choi et al. [20] (2025) Choi et al. proposed a novel ABC ensemble model for skin lesion classification that incorporated the clinically established ABCD rule, which evaluates lesion asymmetry, border, color, and diameter. The model consisted of five distinct blocks, where two blocks learned general image characteristics and the remaining three focused on ABCD-specific lesion features. A weighted soft voting strategy was used for the final classification, with higher weights assigned to the generalization blocks to optimize overall performance. Across 15 experimental configurations, the weighted ABC ensemble achieved 93.26% accuracy and an F1-score of 93.02%, outperforming all individual baseline models. Grad-CAM analysis was applied to assess how each block attended to distinct lesion features, improving the model's interpretability and clinical reliability. The practical contribution of this work is that it bridges clinical dermatological knowledge with deep learning by explicitly embedding ABCD rule criteria into the model architecture. However, the reliance on five separate blocks increases model complexity, and broader evaluation across more diverse and imbalanced datasets would be needed to confirm its generalizability.

Priyadharsini [21] (2026) Priyadharsini proposed HHO-Reg-SA-Net, a novel attention-based optimized network for skin lesion classification that combined deep learning with metaheuristic optimization. The backbone of the model was RegNetY032, enhanced with a Soft Attention Block to improve feature extraction from dermoscopic images. The Harris-Hawks Optimization Algorithm was used to tune hyperparameters, helping the model achieve better generalization without manual configuration. This design addressed the challenge of limited dermatological resources in rural and underserved regions by providing an automated diagnostic tool. The integration of soft attention allowed the network to focus on diagnostically relevant skin regions, reducing the influence of irrelevant background features. The practical contribution is a computationally guided classification framework that improves upon standard transfer learning baselines through optimizer-driven fine-tuning. However, broader multi-dataset validation and class imbalance handling were not extensively addressed in the study.

Gabani et al.[22] (2026) Gabani et al. developed a multimodal ensemble deep learning framework for early skin cancer diagnosis using dermoscopic images from the HAM10000

dataset. The study combined three prominent DCNN architectures, namely DenseNet-201, VGG16, and InceptionV3, through an ensemble strategy evaluated with K-fold cross-validation for reliable performance estimation. The proposed ensemble model achieved a testing accuracy of 97.9%, outperforming all three individual backbone models on the classification task. Grad-CAM was incorporated to improve model interpretability, generating visual explanations that highlight the lesion regions influencing each prediction. The practical contribution of this work is that multi-architecture fusion can significantly improve classification reliability compared to using a single pretrained model. The probabilistic calibration of the ensemble was also analyzed, adding an additional layer of trustworthiness for clinical decision support. However, the study was limited to a single dataset and the computational overhead of running three models simultaneously may limit deployment in resource-constrained settings.

Li et al. [23] (2026) Li et al. proposed SCTFD, a Synthetic Classification Transformer Framework for Dermoscopy designed to address two major challenges in skin lesion classification: data imbalance and high intra-class variance. The framework first applied CN-SMOTE, an encoder-decoder-based nearest sampling synthesis approach, to generate minority class samples and correct the imbalanced class distribution. Feature extraction was then performed by MARD-Net, a Multi-head Attention Residual Dilated Network that combined spatial-channel attention to enhance CNN performance with global sliding window attention to reduce transformer computational complexity. This hybrid design allowed the model to capture both local texture features and broader contextual patterns in dermoscopic images. Evaluated on the HAM10000 dataset, the SCTFD framework achieved state-of-the-art classification performance by effectively handling minority lesion categories. Its practical strength lies in the joint treatment of data synthesis and feature learning within a unified pipeline. However, the complexity of the multi-stage framework may limit its adaptability to smaller clinical datasets without significant preprocessing effort.

Abdallah et al. [24] (2026) Abdallah et al. introduced an advanced deep learning framework that integrated a Scaled Dot Attention Mechanism with a modified UNet architecture for skin cancer detection and segmentation. The SDAM was applied between the encoder and decoder stages of the UNet, allowing the model to prioritize diagnostically relevant lesion regions while suppressing background noise. The model was evaluated on three datasets, including HAM10000, ISIC Preliminary, and PH2 Preliminary, demonstrating strong generalizability across different lesion types and imaging conditions. It achieved Dice scores

ranging from 0.97 to 0.988 and accuracy between 97.8% and 98.3%, outperforming the standard UNet baseline which scored a Dice of 0.85 and 88.4% accuracy. The multi-dataset evaluation is a notable practical contribution, as it addresses the generalization gap that affects models trained on a single-source dataset. The attention-enriched architecture strengthened feature extraction and made the segmentation boundaries more precise and clinically reliable. A limitation noted is that the HAM10000 dataset is biased toward lighter skin tones, which raises fairness concerns that remain unresolved in the current work.

Raza et al. [25] (2026) Raza et al. proposed a multimodal deep learning approach that fused patient metadata with dermoscopic skin lesion images to improve skin cancer detection performance. Three separate models were developed and evaluated: a metadata-only model achieving 85.24% sensitivity, an image-only model achieving 99.72% sensitivity, and a combined fused model that reached 99.66% sensitivity and 74.45% specificity. A majority voting ensemble of all three models was then applied, achieving 99.50% sensitivity and 82.72% specificity, significantly outperforming image-only state-of-the-art approaches. A soft-attention post-processing module was incorporated to generate visual explanations of model decisions, supporting transparency for healthcare professionals. The practical contribution of this study is that combining patient-level clinical metadata with image data can improve specificity without sacrificing sensitivity, which is critical for reducing false positives in skin cancer screening. The work was motivated by growing waiting times for lesion assessment in healthcare systems and the need for faster automated decision support. However, the reliance on structured patient metadata may limit applicability in settings where such data is incomplete or unavailable.

Farsi et al. [26] (2026) Farsi et al. proposed a deep learning model inspired by transformer networks for multi-class skin lesion classification, built on an advanced DenseNet-121 backbone with a novel Feature Reinforcement Block. The preprocessing stage applied Gabor filtering and color channel weighting to enhance lesion visibility and improve input quality before deep feature extraction. The Feature Reinforcement Block, which is the core innovation of the study, combined three components: a wavelet transform to capture local features such as edges and textures, a Multi-Head attention mechanism to focus the model on diagnostically prominent regions, and an LSTM-Similarity Module to strengthen sequential feature relationships. This hybrid design effectively combined the dense connectivity strength of DenseNet-121 with the global context modeling capability of transformer-inspired attention, enabling better discrimination across multiple lesion categories.

2.2 GAP IN EXISTING RESEARCH

Although previous studies have shown major improvements in skin lesion classification, several important research gaps still remain. Many existing works mainly focus on binary classification, where lesions are identified only as malignant or benign, instead of handling a more practical multi-class setting involving several different lesion categories. This reduces their usefulness in real clinical environments where dermatologists must accurately distinguish between multiple skin conditions. In addition, most standalone architectures perform well only in certain feature areas. Models such as EfficientNetV2 are highly effective in capturing fine texture details, while architectures like ConvNeXt are stronger at learning broader spatial patterns. However, very few studies have explored hybrid systems that combine the strengths of different architectures to achieve more balanced feature learning.

Another major limitation in existing research is the lack of interpretability and imbalance handling. Although techniques like Grad-CAM are available, many published models still fail to provide visual explanations for their predictions, making it difficult for clinicians to understand whether the model is focusing on the correct lesion regions. This lack of transparency remains a challenge for clinical adoption. At the same time, class imbalance in datasets such as HAM10000 is not consistently addressed across studies. As a result, some systems may report high overall accuracy while performing poorly on minority lesion classes, which can lead to unreliable clinical outcomes. To overcome these limitations, the proposed project introduces a multi-class classification framework using a hybrid dual-architecture approach along with Grad-CAM visualization to provide both improved performance and more transparent, clinically meaningful predictions.

SUMMARY

This chapter reviewed twenty research studies related to automated skin lesion classification using deep learning techniques. The studies covered various approaches such as CNN models, ensemble learning, attention mechanisms, hybrid architectures, and Vision Transformers using HAM10000 and ISIC datasets. Models including EfficientNet, Xception, DenseNet, and ConvNeXt showed promising performance in lesion classification. However, many existing systems are still limited by binary classification, lack of interpretability, poor handling of class imbalance, and dependence on single architectures. To overcome these limitations, the proposed project introduces a hybrid multi-class framework integrated with Grad-CAM visualization for improved accuracy and transparency.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

- Memory : 8 GB RAM (Minimum)
- CPU : Intel i5 / AMD Ryzen 5 Processor or higher
- GPU : NVIDIA GPU
- File Size : 5 GB
- Storage : 50 GB (Minimum)
- Operating System : Windows 10 / Windows 11 / Linux

3.2 SOFTWARE REQUIREMENTS

- **Language** : Python
- **Tools** : Google Colab
- **Libraries** : NumPy, Pandas, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, Matplotlib
- **Platform** : Google Colab

3.2.1 PROGRAMMING ENVIRONMENT

PYTHON

Python is a high-level, object-oriented programming language widely used for deep learning, artificial intelligence, and medical image analysis. It is simple, flexible, and efficient, making it suitable for implementing skin lesion classification models. In this project, Python is used to develop deep learning architectures such as EfficientNetV2-S, ConvNeXt, DenseNet, and the proposed hybrid model for multi-class skin lesion classification using dermoscopic images.

Python provides easy-to-understand syntax and powerful libraries that support image preprocessing, model training, evaluation, and visualization. It integrates seamlessly with tools like Google Colab and Jupyter Notebook, enabling efficient development and experimentation. Its interpreted nature allows direct execution without compilation and supports cross-platform development.

Furthermore, Python has a vast ecosystem of frameworks and libraries specifically designed for deep learning and computer vision, such as TensorFlow, Keras, and PyTorch.

These libraries offer pre-built functions and models that simplify the implementation of complex neural networks and reduce development time. Python also supports data handling and analysis through libraries like NumPy and Pandas, making it easier to manage large medical image datasets. Overall, Python provides a comprehensive and efficient environment for developing, testing, and deploying advanced deep learning models in healthcare applications.

ADVANTAGES OF PYTHON

- Supports deep learning and scientific computing through libraries such as TensorFlow, Keras, PyTorch, NumPy, and Pandas.
- Open-source and easy to learn for both beginners and advanced users.
- Provides flexibility for implementing image processing and AI-based healthcare applications.
- Large community support and extensive documentation.

DISADVANTAGES OF PYTHON

- Slower execution compared to compiled languages like C and Java.
- High memory usage when handling large image datasets and deep learning models.

3.2.2 DEVELOPMENT PLATFORM

GOOGLE COLAB

Google Colab (Google Collaboratory) is a cloud-based platform widely used for Python programming, deep learning, and medical image analysis. It provides a free and user-friendly environment for developing and executing skin lesion classification models. In this project, Google Colab is used to implement and train models such as EfficientNetV2-S, ConvNeXt, DenseNet, and the hybrid deep learning model on the HAM10000 dataset.

Google Colab can be accessed through a web browser without installation and supports integration with Google Drive for dataset storage. It provides built-in support for deep learning libraries and GPU acceleration, making it suitable for training complex convolutional neural networks efficiently. The availability of pre-installed libraries reduces setup time and allows direct focus on model development and experimentation.

Additionally, Google Colab offers an interactive notebook environment where code, visualizations, and documentation can be combined in a single workspace. This improves readability and helps in tracking experiments, comparing results, and debugging errors. It also supports real-time collaboration, enabling multiple users to work on the same notebook

simultaneously, which is especially useful in academic and research-based projects. Overall, Google Colab serves as a powerful and accessible platform for implementing deep learning solutions in medical image classification tasks.

ADVANTAGES OF GOOGLE COLAB

- Free cloud-based platform accessible from any browser.
- Provides GPU/TPU support for faster deep learning model training.
- Supports collaboration with multiple users in real time.
- Easy integration with Google Drive for dataset handling.
- Eliminates the need for high-end local hardware.

DISADVANTAGES OF GOOGLE COLAB

- Requires a stable internet connection.
- Session time limits may interrupt long training processes.
- Limited RAM and GPU availability in the free version.
- Not ideal for very large-scale datasets or continuous long-duration training.

SUMMARY

The system requirements for this project include a minimum of 8GB RAM, a modern processor such as Intel i5 or AMD Ryzen 5, and optional GPU support to ensure efficient training of deep learning models on medical image datasets. Python is used as the core programming language due to its simplicity, flexibility, and extensive support for deep learning frameworks such as TensorFlow, Keras, and PyTorch, enabling the implementation of advanced architectures like EfficientNetV2-S, ConvNeXt, DenseNet, and a hybrid model for accurate skin lesion classification. Google Colab serves as the primary development platform, providing a cloud-based environment with built-in libraries, GPU/TPU acceleration, and seamless integration with Google Drive for data storage and management. The entire system is developed and executed within this environment without the need for a separate frontend or backend architecture. Although Google Colab has certain limitations such as internet dependency, session timeouts, and restricted resources in the free version, it remains a highly accessible and efficient platform for deep learning applications. Overall, the combination of Python and Google Colab provides a robust, scalable, and practical solution for developing and evaluating a reliable skin lesion classification system in the healthcare domain.

CHAPTER 4

METHODOLOGY

4.1 PROPOSED METHODOLOGY

This project follows a quantitative and experimental approach based on a data-driven methodology, where dermoscopic images are processed and classified using supervised deep learning models. The overall workflow consists of dataset acquisition, preprocessing, data augmentation, feature extraction using convolutional neural networks, classification, performance evaluation, and interpretability analysis using Grad-CAM.

The study adopts a comparative approach by evaluating three individual architectures—EfficientNetV2-S, ConvNeXt, and DenseNet—alongside a proposed hybrid model that combines EfficientNetV2-S and ConvNeXt. Model performance is assessed using standard evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. All models are trained and evaluated under consistent conditions to ensure a fair and unbiased comparison.

4.2 DATASET DESCRIPTION

The primary dataset used in this study is the HAM10000 (Human Against Machine with 10,000 training images) dataset, one of the most widely used and respected benchmarks in dermatological image analysis. The images in this collection were sourced from multiple clinical centers in Austria and Australia, giving the dataset a realistic and diverse character that closely reflects real-world clinical conditions.

All images in the HAM10000 dataset were captured using dermoscopy, a non-invasive technique that allows clinicians to examine subsurface skin structures that are not visible to the naked eye. Dermoscopic imaging reveals critical diagnostic features including pigment networks, vascular patterns, streak formations, and globular structures, providing far richer visual information than standard photography.

In total, the dataset contains 10,015 dermoscopic images distributed across multiple diagnostic categories. This study, four clinically relevant classes are selected for classification:

Table 4.1: Skin Lesion Classes Used in This Study

Class Code	Lesion Type	Clinical Significance
nv	Melanocytic Nevi	Common benign pigmented lesion; largest class in dataset
mel	Melanoma	Aggressive malignant skin cancer with metastatic potential
bcc	Basal Cell Carcinoma	Most common form of skin cancer; locally invasive
akiec	Actinic Keratosis	Pre-malignant lesion; may progress to squamous cell carcinoma

These four lesion types were selected because they are among the most commonly encountered conditions in clinical dermatology, making the classification task both practically relevant and medically meaningful. Each class presents unique visual characteristics variations in asymmetry, border irregularity, color distribution, and texture that serve as key diagnostic indicators and form the basis for the model's learning process.

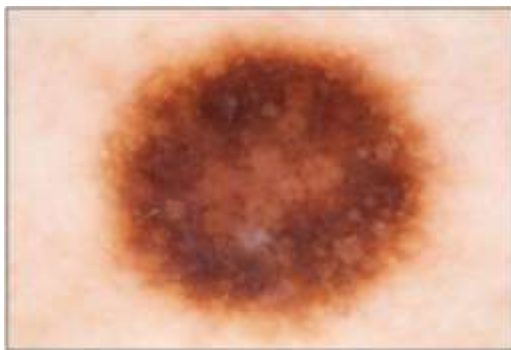
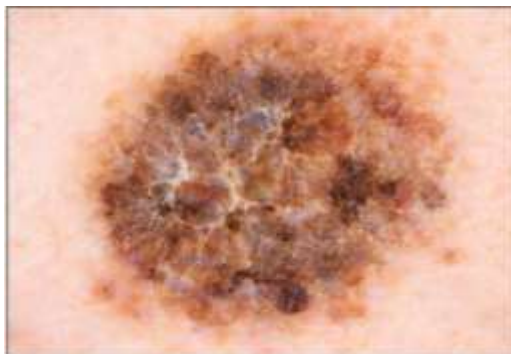
**Figure.4.2.1 Melanocytic Nevi****Figure.4.2.2 Melanoma Cell****Figure.4.2.3. Benign Keratosis****Figure.4.2.4 Basal Cell Carcioma**

Figure 4.2.1–4.2.4 shows sample dermoscopic images from each of the four lesion classes. Even without technical knowledge, the visual differences between the categories are clear. Variations in asymmetry, border irregularity, color distribution, and texture carry important diagnostic value. Deep learning is particularly suitable for this type of analysis because the model can learn these patterns directly from the images without manual feature design. To provide a fair and reliable evaluation, the dataset was divided into three subsets 70% for training, 15% for validation, and 15% for testing.

The training set helps the model learn lesion patterns from the images, the validation set is used to monitor performance and tune the model during training, and the test set is kept fully separate so that the final evaluation reflects the model’s true performance on unseen data. This split helps create a clean and unbiased training environment for accurate skin lesion categorisation. Before any training took place, every image in the dataset was resized to 224×224 pixels. This is the standard input size used by architectures such as EfficientNetV2, DenseNet, and ConvNeXt, because standardizing input dimensions across the dataset keeps processing uniform and removes unnecessary variability before it can affect the models.

All of this, the clinically meaningful class selection, the carefully chosen dataset, the balanced split, and the regular resizing adds up to a preparation pipeline that gives the models a clean and well-structured basis for training. The outcome is a training environment designed to capture the discriminative features needed for accurate and trustworthy skin lesion classification.

4.3 DATA COLLECTION METHOD

The HAM10000 dataset is publicly available through the International Skin Imaging Collaboration (ISIC) Archive, a well-established platform for dermatological image research. The dataset was downloaded and organized for use in this study without any manual image collection. Each image in the dataset is accompanied by ground-truth labels assigned through established clinical procedures, including histopathological examination, dermoscopy follow-up, and expert consensus, ensuring that the labels are medically verified and reliable.

For this study, the dataset was divided into three subsets: 70% for training, 15% for validation, and 15% for testing. This stratified split ensures that each class is proportionally represented across all subsets, preventing any single class from dominating the training or evaluation process. The test set was kept strictly separate from the training process and was used only for final model evaluation.

DATASET IMAGE COUNT SUMMARY

The HAM10000 dataset contains 10,015 total images. For this study, the four selected classes are:

- nv (Melanocytic Nevi) — 6705 the largest and most common class in the dataset
- mel (Melanoma) — 1,113 malignant lesion with high clinical importance
- bcc (Basal Cell Carcinoma) — 514 common skin cancer with visible lesion characteristics
- akiec (Actinic Keratosis) — 327 pre-malignant lesion requiring careful detection

These four classes provide a strong foundation for multi-class skin lesion classification because they represent both benign and malignant conditions that are clinically significant and visually diverse.

These splits ensure that all four lesion classes are represented across training, validation, and testing. Since the dataset is highly imbalanced, especially with nv forming the largest class and akiec the smallest, this distribution helps the model learn more effectively while still preserving a fair evaluation setup.

Training Dataset (Total: 6,062 images)

- nv (Melanocytic Nevi): 4,694 images
- mel (Melanoma): 779 images
- bcc (Basal Cell Carcinoma): 360 images
- akiec (Actinic Keratosis): 229 images

Validation Dataset (Total: 1,299 images)

- nv (Melanocytic Nevi): 1,006 images
- mel (Melanoma): 167 images
- bcc (Basal Cell Carcinoma): 77 images
- akiec (Actinic Keratosis): 49 images

Testing Dataset (Total: 1298 images)

- nv (Melanocytic Nevi): 1,005 images
- mel (Melanoma): 167 images

- bcc (Basal Cell Carcinoma): 77 images
- akiec (Actinic Keratosis): 49 images

4.4 TOOLS AND TECHNIQUES USED

The following tools and frameworks were used in the development of this project:

Table 4.3: Tools and Frameworks Used in the Project

Tool / Framework	Purpose
Python 3.x	Primary programming language
PyTorch	Deep learning framework for model development and training
TensorFlow / Keras	Used in EfficientNetV2 model training pipeline
timm (PyTorch Image Models)	Pretrained model loading for ConvNeXt and EfficientNetV2
scikit-learn	Evaluation metrics: accuracy, classification report, confusion matrix
NumPy & Pandas	Data handling and numerical computation
Matplotlib & Seaborn	Visualization of training curves, confusion matrix, and ROC curves
Grad-CAM (pytorch-gradcam)	Model interpretability and activation heatmap generation
Google Colab / GPU	Training environment with GPU acceleration
ISIC Archive	Dataset source for HAM10000

Table 4.3 presents the tools and frameworks used in developing the proposed skin lesion classification system. Python 3.x was used as the main programming language, while PyTorch and TensorFlow/Keras supported model development and training. The timm library was used for loading pretrained models such as EfficientNetV2 and ConvNeXt. Libraries including NumPy, Pandas, scikit-learn, Matplotlib, and Seaborn were used for data processing, evaluation, and visualization. The experiments were conducted in Google Colab with GPU support using the HAM10000 dataset from the ISIC Archive.

4.5 TECHNIQUES IN DEEP LEARNING

Step 1: Data Preprocessing

Before training, all dermoscopic images were resized to a uniform resolution of 224×224 pixels, which is the standard input dimension required by EfficientNetV2, ConvNeXt, and DenseNet architectures. Pixel values were then normalized to the range $[0, 1]$ by dividing by 255, which stabilizes gradient flow during optimization and reduces the influence of lighting and contrast differences between images. Additional artifact mitigation was applied to address common issues in dermoscopic images such as hair occlusion, uneven illumination, and background noise.

Step 2: Data Augmentation

To improve model generalization and reduce the risk of overfitting, data augmentation was applied to the training set. Augmentation operations included random horizontal and vertical flips, random rotations, zooming, translation shifts, and random cropping. These transformations introduce controlled variety into the training data, simulating the natural differences in lesion appearance, orientation, and scale that would be encountered across real patients and imaging sessions. The validation and test sets were not augmented, ensuring that evaluation reflects performance on unmodified images.

Step 3: Feature Extraction

Three deep learning architectures were used for feature extraction: EfficientNetV2-S, ConvNeXt, and DenseNet. Each architecture was initialized with pretrained ImageNet weights and then fine-tuned on the HAM10000 dataset. Pretrained weights provide a strong initialization that accelerates convergence and improves final performance, particularly when the available training data is limited relative to model complexity.

Step 4: Hybrid Feature Fusion

The proposed hybrid model feeds each input image simultaneously into two parallel branches: EfficientNetV2 and ConvNeXt. Each branch independently extracts its own set of feature representations. Global average pooling is then applied to both branches to reduce spatial dimensions into compact feature vectors. These vectors are concatenated into a single unified representation that captures both the fine-grained detail from EfficientNetV2 and the high-level spatial structure from ConvNeXt.

Step 5: Classification

The combined feature vector passes through fully connected layers where class-specific patterns are learned. A dropout layer is applied within this stage to prevent overfitting and improve generalization. A final softmax layer converts the output into probability scores across the four target classes, with the class receiving the highest probability selected as the model's prediction.

Step 6: Interpretability with Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to generate visual heatmaps that highlight the regions of each dermoscopic image most influential in driving the model's prediction. These heatmaps are overlaid on the original images and provide clinicians with a visual explanation of the model's attention, confirming whether the model is focusing on diagnostically relevant lesion features.

4.6 IMPLEMENTATION METHODOLOGY

In this study, four deep learning models—EfficientNetV2, ConvNeXt, DenseNet, and a hybrid model combining EfficientNetV2 with ConvNeXt—were implemented and evaluated for multi-class skin lesion classification using dermoscopic images from the HAM10000 dataset. Each model was trained to learn visual patterns from skin lesion images and classify them into four clinically relevant categories: melanocytic nevi, melanoma, basal cell carcinoma, and actinic keratosis. The comparison between these models helped assess how well each architecture captured both fine-grained lesion details and broader spatial features, while also considering performance, robustness, and computational efficiency.

Based on the experimental results, a hybrid dual-architecture model was developed by combining EfficientNetV2 and ConvNeXt to improve classification performance. This hybrid approach takes advantage of EfficientNetV2's strong feature extraction capability and ConvNeXt's ability to learn rich spatial representations. By integrating the strengths of both models, the proposed system produces more balanced and reliable predictions across all four lesion classes. In addition, Grad-CAM was incorporated to improve interpretability by visually highlighting the lesion regions that influenced the model's decisions. This makes the final system not only accurate but also more transparent and clinically meaningful for skin lesion diagnosis. Furthermore, the hybrid framework reduces the limitations of relying on a single deep learning model by combining complementary feature learning strategies.

4.7 MODEL ARCHITECTURE

4.7.1 EfficientNetV2-S

EfficientNetV2-S is a state-of-the-art CNN architecture built on the principle of compound scaling, which simultaneously adjusts network depth, width, and input resolution in a coordinated manner. Unlike conventional architectures that scale these dimensions independently, EfficientNetV2 achieves a more balanced and efficient model that delivers high classification performance without an excessive increase in parameter count.

The architecture is built on two types of core building blocks: MBConv layers, which use depthwise separable convolutions to reduce computational cost while preserving feature quality, and Fused-MBConv layers, which combine standard convolution with expansion operations for faster training in early network stages. Together, these blocks allow EfficientNetV2 to learn both fine-grained textures and broader visual patterns efficiently. For dermoscopic image analysis, this translates into a model capable of detecting the subtle color variations, pigmentation distributions, and boundary features that are critical for accurate lesion classification.

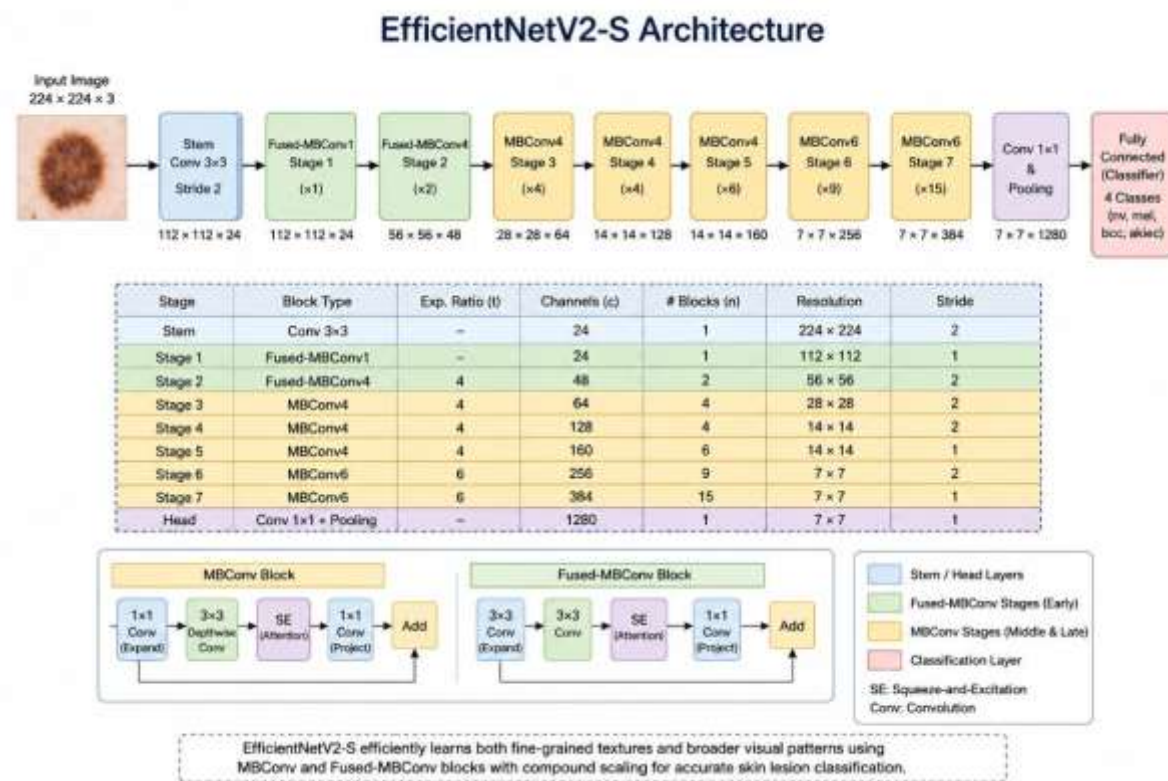


Figure 4.7.1.1 EfficientNetV2-S Architecture

Figure 4.7.1.1 illustrates the overall structure of the EfficientNetV2-S architecture used in this study for skin lesion classification. The model begins with an input dermoscopic image of size $224 \times 224 \times 3$ and passes it through a stem convolution layer for initial feature extraction. After that, the network progresses through a sequence of Fused-MBConv and MBConv blocks, where early stages focus on efficient low-level feature learning and deeper stages capture richer and more abstract lesion patterns.

The feature map resolution gradually decreases while the number of channels increases, allowing the model to learn more complex visual representations with better efficiency. At the end of the network, the extracted features are processed through a 1×1 convolution and pooling layer before being sent to the final classifier, which predicts one of the four skin lesion classes: nv, mel, bcc, and akiec. The diagram also highlights the internal composition of MBConv and Fused-MBConv blocks, showing how the architecture balances accuracy and computational efficiency for dermoscopic image analysis.

4.7.2 ConvNeXt

ConvNeXt is a modernized CNN architecture inspired by the design principles of Vision Transformers (ViTs). Rather than replacing convolutional operations with transformer attention mechanisms, ConvNeXt retains the efficiency of CNNs while adopting several key design improvements: larger convolutional kernels, layer normalization instead of batch normalization, a stage-wise hierarchical structure, and improved residual connections.

The residual connections in ConvNeXt are expressed as $Y = F(X) + X$, where X is the input feature map, $F(X)$ is the residual function learned by the network, and Y is the resulting output. This formulation keeps gradient flow stable during backpropagation and prevents performance degradation in deeper layers. ConvNeXt is particularly effective at capturing high-level spatial structures, making it a strong complement to EfficientNetV2 in the proposed hybrid framework.

In addition to its strong feature extraction capability, ConvNeXt provides better scalability and computational efficiency for deep image classification tasks. The architecture is designed with depthwise separable convolutions and optimized network blocks, allowing it to learn complex visual representations while maintaining lower computational overhead compared to many transformer-based models. Its hierarchical feature learning strategy enables the network to progressively capture both local and global contextual information from dermoscopic images, which is important for distinguishing visually similar skin lesions.

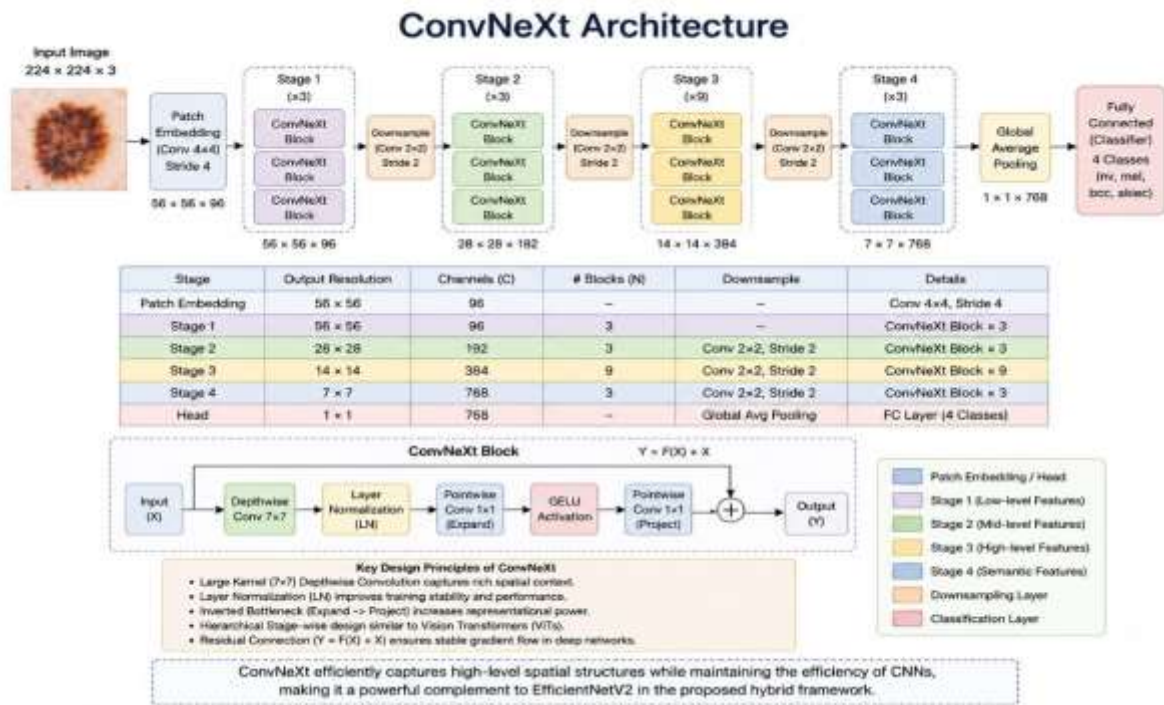


Figure 4.7.2.1 ConvNeXt Architecture

Figure 4.7.2.1 illustrates the overall architecture of the ConvNeXt model used for skin lesion classification in this study. The model begins with a $224 \times 224 \times 3$ input image, followed by a patch embedding layer that converts the image into feature representations. The architecture is organized into multiple hierarchical stages consisting of ConvNeXt blocks, where the spatial resolution gradually decreases while the number of feature channels increases. Each ConvNeXt block uses depthwise convolution, layer normalization, GELU activation, and residual connections to improve feature learning and maintain stable gradient flow. The residual formulation $Y = F(X) + X$ helps preserve important information across deeper layers of the network. Finally, the extracted features are passed through a global average pooling layer and a fully connected classifier to predict the four skin lesion categories: nv, mel, bcc, and akiec.

4.7.3 DenseNet

DenseNet (Densely Connected Convolutional Network) adopts a distinctive connectivity pattern where every layer within a dense block receives direct input from all preceding layers: $x_i = H_i([x_0, x_1, \dots, x_{i-1}])$. This design naturally encourages feature reuse, reduces redundancy, and makes the network far more parameter-efficient than traditional architectures. DenseNet is also less susceptible to the vanishing gradient problem, as gradients can flow directly across

multiple pathways during backpropagation. In this study, DenseNet serves as an additional baseline architecture for performance comparison.

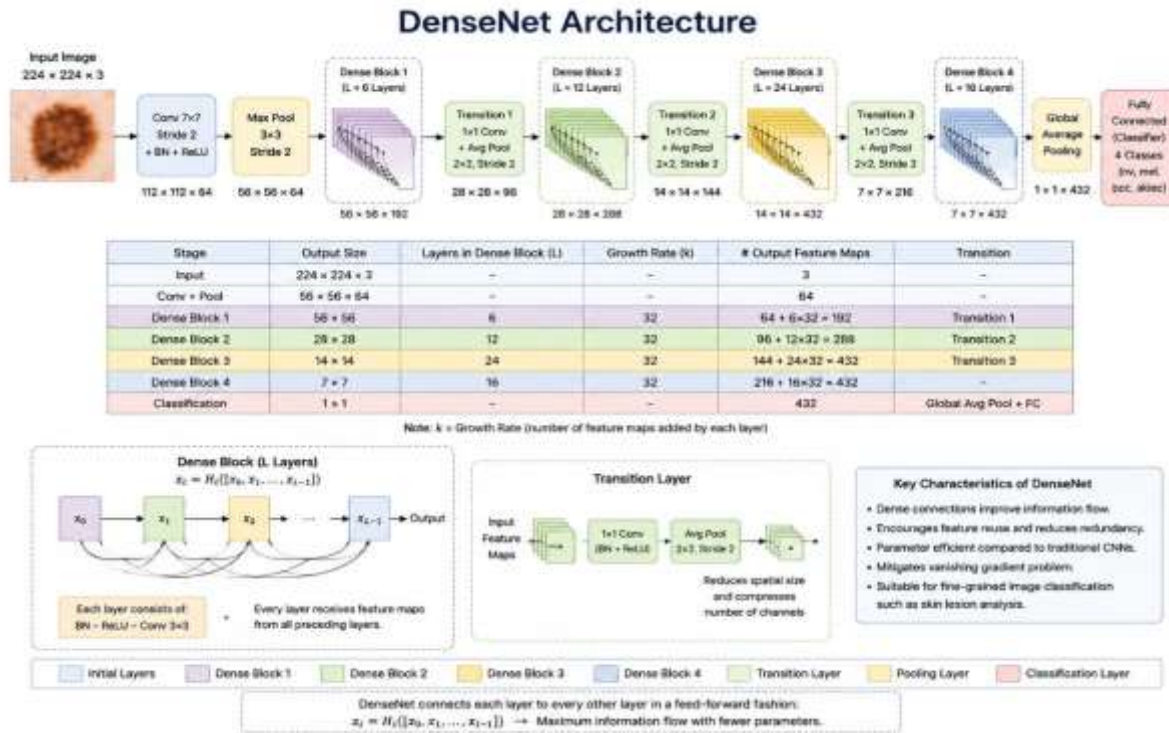


Figure 4.7.3.1 DenseNet

Figure 4.7.3.1 illustrates the architecture of the DenseNet model used in this study for skin lesion classification. The model begins with an input dermoscopic image followed by convolution and pooling layers for initial feature extraction. DenseNet is organized into multiple dense blocks, where each layer receives feature maps from all preceding layers, enabling efficient feature reuse and improved information flow throughout the network. Between dense blocks, transition layers are used to reduce feature map size and computational complexity through convolution and average pooling operations. Finally, the extracted features are passed through a global average pooling layer and a fully connected classifier to predict the four skin lesion categories: nv, mel, bcc, and akiec.

4.7.4 PROPOSED HYBRID MODEL

The hybrid model combines EfficientNetV2 and ConvNeXt in a dual-branch parallel architecture. The two branches operate independently, with each extracting a distinct set of feature representations from the same input image. The feature extraction processes are formalized as:

$$F_{\text{EfficientNetV2}} = f_{\text{EfficientNetV2}}(X)$$

$$F_ConvNeXt = f_ConvNeXt(X)$$

The hybrid feature vector is obtained through concatenation:

$$F_hybrid = [F_EfficientNetV2 \oplus F_ConvNeXt]$$

The final classification output is computed as:

$$y = \text{Softmax}(W \cdot F_hybrid + b)$$

where W and b represent the learnable parameters of the classification layer. The dual-branch design ensures that neither the fine-grained sensitivity of EfficientNetV2 nor the structural awareness of ConvNeXt is lost in the final prediction, resulting in a more robust and balanced classification system.

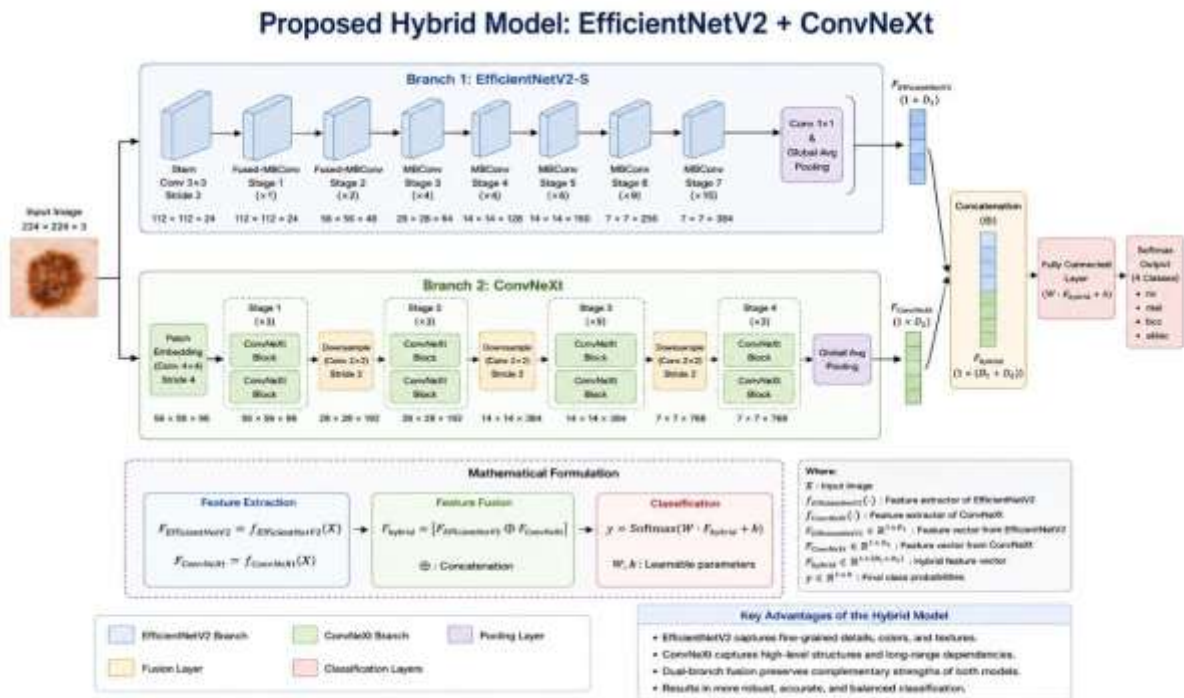


Figure 4.7.4.1 Proposed Hybrid Model: EfficientNetV2 + ConvNeXt

Figure 4.7.4.1 illustrates the proposed hybrid architecture that combines EfficientNetV2-S and ConvNeXt in a dual-branch parallel framework for skin lesion classification. The same dermoscopic input image is simultaneously processed by both branches, where EfficientNetV2 focuses on extracting fine-grained textures and color details, while ConvNeXt captures broader spatial and structural patterns. The feature vectors generated from both models are fused through a concatenation layer to create a richer hybrid representation. The combined features are then passed through a fully connected layer followed by a Softmax classifier to predict the four lesion categories: nv, mel, bcc, and akiec. This dual-branch design improves feature diversity and produces a more robust and balanced classification system.

4.7.4.2 WORK FLOW OF THE PROPOSED ARCHITECTURE

The proposed hybrid architecture begins by preprocessing the dermoscopic images from the HAM10000 dataset to ensure uniformity and improve model performance. The input images are first resized to 224×224 pixels and normalized so that all samples have a consistent format for training. After preprocessing, the images are fed into two parallel feature extraction branches: EfficientNetV2-S and ConvNeXt. EfficientNetV2-S focuses on learning fine-grained lesion details such as texture, color variation, and border irregularity, while ConvNeXt captures broader spatial structures and higher-level visual patterns. The feature representations obtained from both branches are then extracted separately and passed to a concatenation layer, where they are merged into a single hybrid feature vector.

This combined representation preserves the complementary strengths of both networks and provides a more complete description of the lesion image. The fused feature vector is then forwarded to the fully connected layer, where the final classification is performed using the Softmax function. The output class corresponds to one of the four skin lesion categories: melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec).

During training, the model learns to minimize classification loss by updating the weights of both branches and the classifier. To improve interpretability, Grad-CAM is used after prediction to highlight the important lesion regions that influenced the model's decision. This workflow makes the proposed system both accurate and clinically meaningful for skin lesion classification.

ALGORITHM

Step 1: Input dermoscopic images from the HAM10000 dataset

Step 2: Apply preprocessing by resizing images to 224×224 and normalizing pixel values

Step 3: Feed the processed images into two parallel branches: EfficientNetV2-S and ConvNeXt

Step 4: Extract fine-grained features using EfficientNetV2-S

Step 5: Extract structural and spatial features using ConvNeXt

Step 6: Obtain feature vectors from both branches

Step 7: Concatenate the extracted feature vectors into a hybrid representation

Step 8: Pass the fused features to the fully connected layer

Step 9: Apply Softmax to classify the image into one of the four lesion classes

Step 10: Compute the loss and update the model weights during training

Step 11: Use Grad-CAM to visualize the important regions influencing the prediction

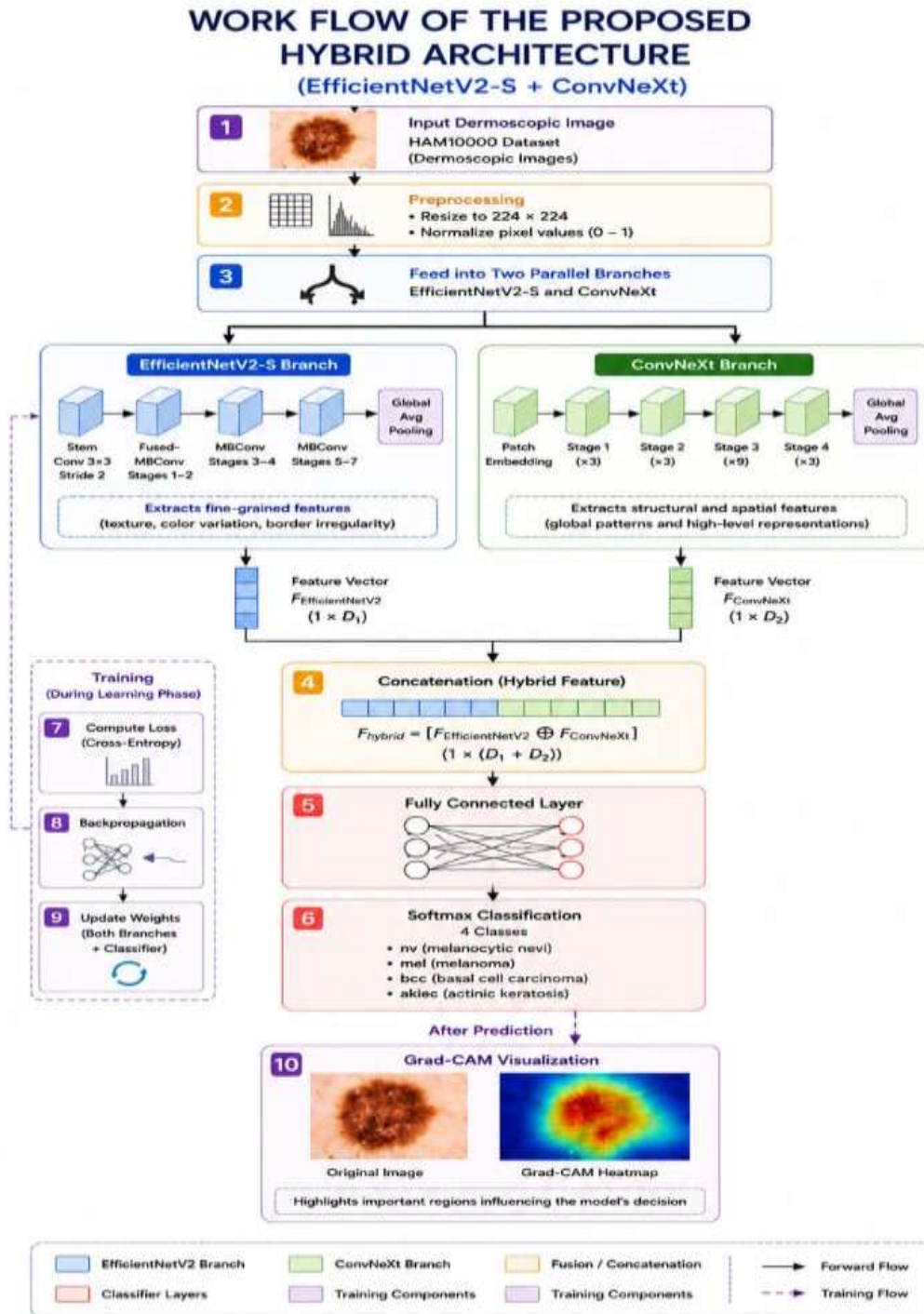


Figure 4.7.4.2 Workflow of the Proposed Hybrid Architecture

Figure 4.7.4.2 illustrates the workflow of the proposed hybrid architecture for multi-class skin lesion classification using dermoscopic images from the HAM10000 dataset. The input images are first preprocessed through resizing and normalization before being passed into two parallel feature extraction branches: EfficientNetV2-S and ConvNeXt. EfficientNetV2-S extracts fine-grained lesion details such as texture and color variation, while ConvNeXt captures broader structural and spatial patterns. The extracted features from both branches are concatenated to form a hybrid feature representation, which is then passed through a fully connected layer and Softmax classifier to predict the four lesion classes. After classification, Grad-CAM visualization is applied to highlight the important lesion regions that influenced the model's prediction.

After feature extraction, the outputs from both branches are fused using a concatenation layer to form a single hybrid feature vector containing detailed and structural information from the lesion images. This fused representation is then forwarded to a fully connected layer, where the Softmax classifier predicts the lesion category among melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec). During the training phase, the model continuously updates its parameters by minimizing the classification loss, thereby improving prediction accuracy and generalization capability. To further enhance clinical interpretability, Grad-CAM visualization is applied after prediction to identify and highlight the important lesion regions that influenced the model's decision-making process. This not only improves transparency in diagnosis but also helps medical professionals understand the reasoning behind the classification results, making the proposed system both effective and clinically meaningful for automated skin lesion analysis.

SUMMARY

This chapter described the complete methodology of the project. The study uses the HAM10000 dataset containing 10,015 dermoscopic images classified into four categories: melanocytic nevi, melanoma, basal cell carcinoma, and actinic keratosis. Data was split 70/15/15 for training, validation, and testing respectively. Preprocessing involved resizing to 224×224 pixels, normalization, and augmentation. Three CNN architectures were implemented: EfficientNetV2-S, ConvNeXt, and DenseNet. A hybrid model was proposed by combining EfficientNetV2 and ConvNeXt through feature concatenation. All models were trained using the Adam optimizer with a learning rate of 0.0001 for 30 epochs. Grad-CAM was incorporated for visual interpretability of model predictions. The framework more suitable for practical medical and clinical decision-support applications.

CHAPTER 5

RESULTS AND DISCUSSION

At present, skin cancer, especially melanoma, is a major concern in healthcare because early and accurate detection is essential for reducing patient risk and improving treatment outcomes. In this work, dermoscopic image data from the HAM10000 dataset is used and implemented in Google Colab using deep learning techniques. The analysis focuses on improving classification accuracy and model performance by utilizing advanced architectures such as EfficientNetV2-S, ConvNeXt, DenseNet, and a hybrid model combining EfficientNetV2-S with ConvNeXt. The proposed system aims to provide a robust and reliable solution for skin lesion diagnosis.

5.1 PERFORMANCE METRICS

1. Accuracy

In the skin lesion classification project, Accuracy measures the overall ability of the model to correctly classify dermoscopic images into their respective lesion categories, such as nv, mel, bcc, and akiec. It represents the proportion of correctly predicted images out of all images in the dataset. A higher accuracy indicates that the model can effectively identify both common and minority lesion classes. However, when the dataset is imbalanced, accuracy alone may not fully reflect the true performance of the model.

Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Where:

- TP (True Positive): Lesion images correctly classified as the target class
- TN (True Negative): Non-target lesion images correctly classified as not belonging to that class
- FP (False Positive): Images incorrectly classified as the target class
- FN (False Negative): Target class images incorrectly classified as another class

2. Precision

In the skin lesion project, Precision measures how accurately the model predicts a particular lesion class. It indicates the proportion of images predicted as a specific class that

are actually correct. High precision is important in skin lesion classification because it reduces false alarms and helps avoid incorrectly labeling a benign lesion as malignant.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

3. Recall

Recall measures the model's ability to correctly identify all images belonging to a particular lesion class. It shows how effectively the model detects actual samples of each category. High recall is especially important in skin cancer analysis because missing a malignant lesion may delay diagnosis and treatment.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4. F1-Score

In the skin lesion classification project, F1-Score provides a balanced evaluation of the model by combining both precision and recall into a single metric. It is particularly useful when the dataset contains class imbalance, where some lesion categories have far fewer samples than others. A high F1-Score indicates that the model performs well while maintaining a balance between false positives and false negatives.

Formula:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

5. ROC Curve (Receiver Operating Characteristic Curve)

In the skin lesion classification project, the ROC Curve is used to evaluate the performance of the deep learning model in distinguishing between different lesion classes. It shows the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) at different threshold values. A curve closer to the top-left corner indicates better performance.

The performance of the ROC curve is commonly summarized using the Area Under the Curve (AUC). A higher AUC value indicates stronger discrimination between skin lesion classes and better classification capability. The classification report for the proposed models shows strong results, with EfficientNetV2-S achieving 0.954334 accuracy, while the hybrid model and DenseNet also provide useful comparison results for multi-class analysis.

Formula for True Positive Rate (TPR):

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

Formula for False Positive Rate (FPR):

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

5.2 CLASSIFICATION REPORT**1. EfficientNetV2-S****Table 5.2.1 Classification Report for EfficientNetV2-S**

Class / Metric	Precision	Recall	F1-Score	Support
Class 0	0.967488	0.982983	0.975174	999
Class 1	0.891026	0.832335	0.860681	167
Class 2	0.923077	0.935065	0.929032	77
Class 3	0.930233	0.816327	0.869565	49
Accuracy	0.954334	0.954334	0.954334	0.954334
Macro Average	0.927956	0.891677	0.908613	1292
Weighted Average	0.953545	0.954334	0.953620	1292

The EfficientNetV2-S model demonstrates strong and consistent performance in classifying the four skin lesion categories. It achieves a high overall accuracy of 0.954334, which shows that the model is highly capable of learning the visual patterns present in dermoscopic images. The results also indicate that the model performs especially well on the majority class, while still maintaining reasonable precision and recall for the smaller classes. This is important because the dataset is imbalanced, and many models tend to favor the larger class while performing poorly on minority categories. In this case, EfficientNetV2-S shows a good balance between overall correctness and class-wise reliability, making it a strong model for multi-class skin lesion classification.

2. DenseNet

The performance of the DenseNet model was evaluated using precision, recall, F1-score, and overall accuracy metrics on the HAM10000 skin lesion dataset. DenseNet is a densely connected convolutional neural network architecture in which each layer receives feature information from all preceding layers, enabling efficient feature reuse and improved gradient propagation. This design helps the model learn both low-level and high-level dermoscopic features effectively while reducing redundancy in feature learning. The model

was trained to classify four skin lesion categories: benign nevus, melanoma, keratinocytic lesions, and dermatofibroma.

Table 5.2.2 Classification Report for DenseNet

Class / Metric	Precision	Recall	F1-Score	Support
Benign Nevus	0.93	0.97	0.95	1341
Melanoma	0.82	0.62	0.71	223
Keratinocytic Lesions	0.86	0.87	0.87	388
Dermatofibroma	0.89	0.78	0.83	51
Accuracy	—	—	0.91	2003
Macro Average	0.88	0.81	0.84	2003
Weighted Average	0.90	0.91	0.90	2003

The DenseNet model serves as a useful baseline for comparison in this study. Its overall accuracy is 0.91, which is lower than EfficientNetV2-S, but still shows that the architecture is able to capture meaningful features from the dermoscopic images. The model performs well on the larger class, where more training examples are available, but its performance drops slightly for the minority classes. This suggests that DenseNet is more affected by class imbalance and may have more difficulty distinguishing visually similar or less represented lesion types. Even so, the model provides valuable insight into how a traditional densely connected architecture performs in comparison with the newer EfficientNet-based model. This highlights the importance of incorporating techniques such as data augmentation or class balancing to further improve DenseNet's performance on underrepresented classes.

3. HYBRID MODEL

Figure 5.2.3 Class-wise Performance of the Proposed Hybrid Architecture

Class	Accuracy	Sensitivity	Specificity
nv	0.942263	0.978374	0.818414
mel	0.945150	0.726457	0.977469
bcc	0.986721	0.873786	0.993861
akiec	0.988453	0.753846	0.997600

Table 5.2.4 Classification Report for Proposed Hybrid Model

Class / Metric	Precision	Recall	F1-Score	Support
Class 0	0.95	0.98	0.96	1341
Class 1	0.83	0.73	0.77	223
Class 2	0.90	0.87	0.89	103
Class 3	0.92	0.75	0.83	65
Accuracy	—	—	0.93	1732
Macro Average	0.90	0.83	0.86	1732
Weighted Average	0.93	0.93	0.93	1732

The proposed hybrid model combines the strengths of EfficientNetV2-S and ConvNeXt to improve classification performance across all four lesion classes. Its overall accuracy of 0.93 shows that the model produces reliable predictions and maintains a balanced performance across the dataset. The hybrid approach is especially useful because EfficientNetV2-S captures fine-grained lesion details such as color and texture, while ConvNeXt contributes stronger structural and spatial feature learning. As a result, the combined model achieves more stable class-wise results and handles the difficult minority classes better than a single architecture alone. This makes the hybrid model a practical and robust choice for multi-class skin lesion diagnosis.

5.3 COMPARISON OF TRAINED MODELS

A side-by-side comparison of the four models reveals important architectural and performance differences that go beyond raw accuracy scores. DenseNet earns its consistent performance through its dense inter-layer connectivity, which promotes feature reuse and keeps gradients flowing steadily during backpropagation. However, its relatively lower accuracy compared to the other models reflects its limitations in capturing the more complex and subtle lesion features that dermoscopic classification demands.

EfficientNetV2-S leads all models in overall accuracy, and its compound scaling approach is the key factor. By simultaneously scaling network depth, width, and resolution, EfficientNetV2 reaches a level of feature extraction precision that independently scaled

architectures cannot match. The model excels at detecting the fine-grained visual cues—subtle color shifts, texture patterns, and local pigmentation structures—that are most critical for accurate skin lesion identification.

ConvNeXt achieves 95% accuracy through a modernized convolutional design that captures richer and more generalizable representations than traditional CNNs. Its larger kernel sizes and improved normalization strategy allow it to pick up on complex dermatological patterns, including irregular pigmentation, lesion boundary definition, and textural variations. These qualities make ConvNeXt a powerful complement to EfficientNetV2 in the hybrid framework.

The hybrid model, while sitting at 94% overall accuracy, demonstrates its true value through consistent and reliable performance across all four lesion categories. By fusing the complementary feature representations from both branches, the hybrid framework reduces class-specific bias and handles visually similar lesion types more reliably than either individual model. This kind of balanced, consistent performance is what matters most in clinical deployment, where reliable classification across every class—not just the most common ones—is essential.

5.4 CONFUSION MATRIX ANALYSIS

The confusion matrix provides a detailed breakdown of classification outcomes at the class level, revealing both where the model performs well and where it is most likely to make errors. In this study, the matrix spans all four target classes: melanocytic nevi (nv), melanoma (mel), basal cell carcinoma (bcc), and actinic keratosis (akiec). High values along the diagonal confirm that the model correctly classifies the large majority of dermoscopic images across all categories. The nevus (nv) class demonstrates the highest rate of correct predictions, which is consistent with its proportionally larger representation in the dataset, giving the model more training examples from which to learn its distinguishing features.

Off-diagonal entries—which represent misclassifications—are sparse overall. The most notable misclassification pattern occurs between melanoma (mel) and actinic keratosis (akiec), two lesion types that share similar visual characteristics including overlapping pigmentation patterns, irregular texture, and comparable boundary structures. Despite these challenges, the proposed hybrid architecture maintains strong classification consistency across all lesion categories by effectively combining fine-grained and structural feature representations.

The confusion matrix therefore demonstrates the robustness of the model and highlights its capability to reduce classification errors in complex multi-class skin lesion diagnosis tasks. This type of confusion is well-documented in the dermoscopy literature and reflects a genuine diagnostic challenge even for experienced clinicians. The fact that misclassifications concentrate in this clinically understandable pattern, rather than appearing randomly across all classes, indicates that the model has learned meaningful and medically relevant feature representations.

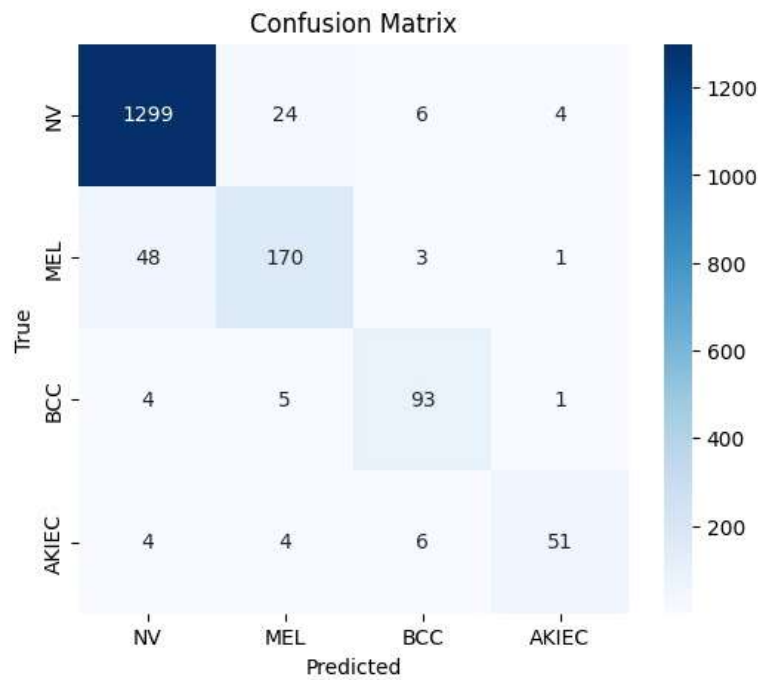


Figure 5.4.1 Confusion Matrix

Figure 5.4.1 the confusion matrix illustrates the classification performance of the proposed deep learning model across four skin lesion classes: NV, MEL, BCC, and AKIEC. Most predictions are concentrated along the diagonal, indicating a high number of correctly classified samples. The NV class achieved the highest correct prediction rate, while a few misclassifications were observed mainly between MEL and AKIEC due to their similar visual characteristics. Overall, the matrix demonstrates the effectiveness of the model in multi-class skin lesion classification.

The confusion matrix also demonstrates that the proposed model maintains balanced classification performance across both majority and minority lesion categories. Minimal confusion between NV, BCC, and AKIEC indicates that the extracted deep learning features are sufficiently distinctive for accurate lesion separation.

5.5 ROC CURVE AND AUC ANALYSIS

The Receiver Operating Characteristic (ROC) curve provides a threshold-independent view of model performance, tracing the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) across all possible decision thresholds. The Area Under the Curve (AUC) summarizes this performance into a single value, where a score closer to 1.0 indicates stronger and more reliable class discrimination.

In this study, high AUC values were recorded for all four lesion classes. Melanocytic nevi (nv) achieved an AUC of 0.98, and melanoma (mel) achieved 0.97, both confirming strong and consistent discriminative capability even as classification thresholds shift. Basal cell carcinoma (bcc) and actinic keratosis (akiec) each achieved an AUC of 1.00, indicating that the model separates these classes from all others with virtually no overlap across the full threshold range.

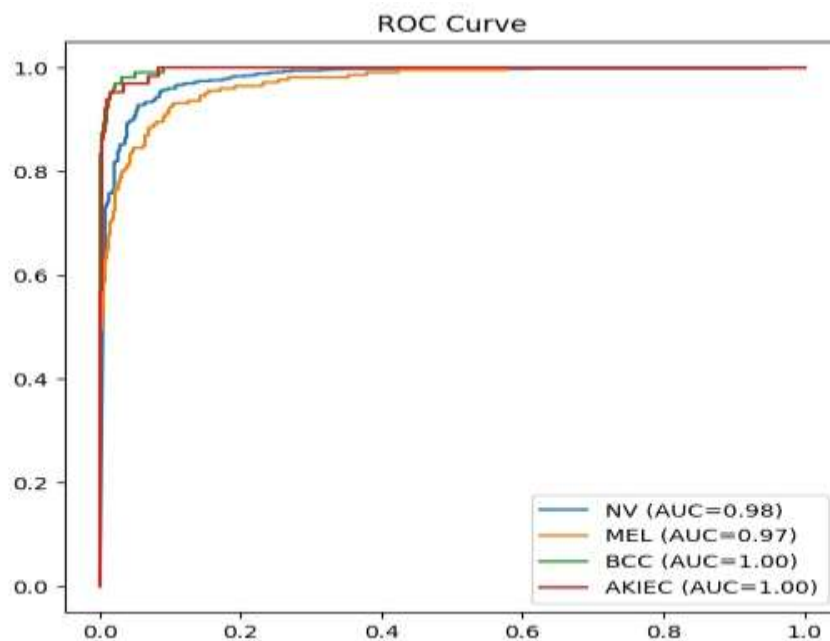


Figure 5.5.1 ROC Curve and AUC Analysis

Figure 5.5.1 the ROC curve illustrates the classification performance of the proposed model across different decision thresholds for all four skin lesion classes. All curves are positioned close to the upper-left corner, indicating high true positive rates with low false positive rates. The model achieved excellent AUC scores, with NV and MEL obtaining 0.98 and 0.97 respectively, while BCC and AKIEC achieved a perfect score of 1.00. These results confirm the strong discriminative capability and reliability of the model in multi-class skin lesion classification.

Table 5.2.5 AUC Scores for Each Lesion Class

Lesion Class	AUC Score
Melanocytic Nevi (nv)	0.98
Melanoma (mel)	0.97
Basal Cell Carcinoma (bcc)	1.00
Actinic Keratosis (akiec)	1.00

All four ROC curves are positioned close to the upper-left corner of the graph, which is the region associated with high true positive rates and low false positive rates—the ideal classification behavior. The uniformly high AUC values across all classes confirm that the model has developed genuinely discriminative representations for each lesion type, and that these representations remain effective regardless of the decision threshold applied.

5.6 TRAINING AND VALIDATION CURVES

The training and validation curves provide a visual account of how each model learned over the course of 30 epochs. For all models, accuracy increased steadily across epochs and approached a plateau toward the end of training, reflecting stable and consistent learning. Validation accuracy followed a similar trajectory and converged to around 93–94% for the hybrid model, confirming that the model generalizes well to unseen data.

The proximity of the training and validation accuracy curves is particularly meaningful. A large gap between the two would indicate overfitting, where the model has memorized the training data rather than learning transferable patterns. The small gap observed here confirms that the model has developed rich, meaningful feature representations rather than simply fitting to noise in the training set.

The loss curves further reinforce this interpretation. Training loss decreased consistently across all epochs, and validation loss remained relatively stable without any significant upward trend after the initial convergence phase. The absence of divergence between training and validation loss confirms that regularization strategies—including dropout and data augmentation—were effective in maintaining generalization throughout training.

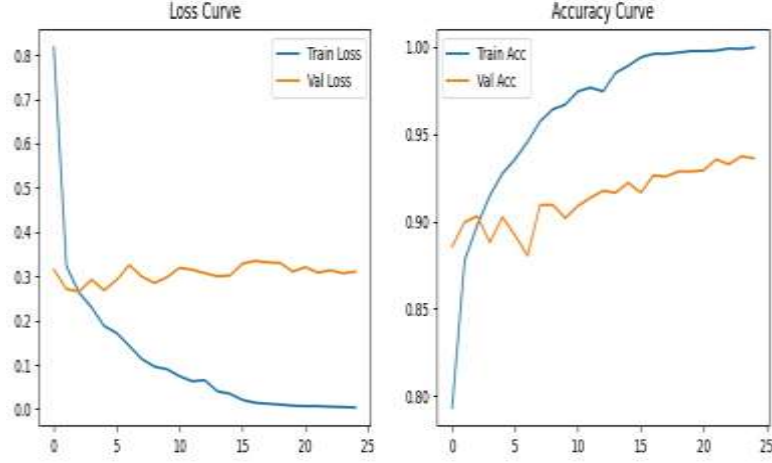


Figure 5.6.1. Training and Validation Curve

Figure 5.6.1 the training and validation curves illustrate the learning performance of the proposed deep learning model over multiple epochs. Both accuracy curves increase steadily and remain closely aligned, indicating stable learning and good generalization capability. Similarly, the training and validation loss curves decrease consistently without significant divergence, showing minimal overfitting during training. Overall, the curves confirm the effectiveness and reliability of the model for skin lesion classification.

5.7 GRAD-CAM VISUALIZATION RESULTS

To improve the transparency and clinical trustworthiness of the proposed framework, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to generate visual heatmaps for individual dermoscopic image predictions. Grad-CAM computes the importance weights for each spatial region of the final convolutional feature map with respect to the predicted class, and generates a heatmap that highlights the areas most influential in driving that prediction.

The Grad-CAM activation map is computed as:

$$L^c = \text{ReLU}(\sum_k \alpha_k^c A_k)$$

where A_k represents the feature maps from the final convolutional layer and α_k^c denotes the importance weight assigned to the k -th feature map for class c . The ReLU operation filters out regions that work against the target class, ensuring that only areas genuinely driving the prediction are highlighted.

The resulting heatmaps consistently highlighted the central lesion region, with warmer colors (red and yellow) concentrated over clinically significant features such as irregular pigmentation, lesion boundary irregularities, and textural variations. These are precisely the features that experienced dermatologists use when evaluating dermoscopic images, providing strong confirmation that the model has learned diagnostically meaningful representations rather than relying on background artifacts or irrelevant image regions.

The inclusion of Grad-CAM visualization enhances the system's overall transparency and serves as a form of indirect validation for model reliability. A classification system that can both produce accurate predictions and explain them visually is significantly more likely to earn clinician trust and acceptance in real-world deployment.

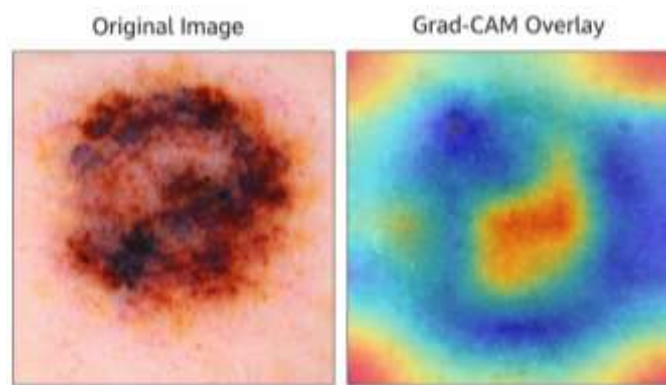


Figure 5.7.1 Grad-CAM Visualization Results

Figure 5.7.1 the Grad-CAM visualization highlights the important regions of the dermoscopic image that influenced the model's prediction. Warmer colors such as red and yellow indicate areas receiving higher attention from the network, mainly concentrated around the lesion region. The activation map confirms that the model focuses on clinically relevant features including irregular pigmentation and lesion boundaries. This visualization improves the interpretability and reliability of the proposed skin lesion classification system.

5.8 DISCUSSION OF FINDINGS

The results of this study demonstrate several important findings that are worth discussing in detail. First, EfficientNetV2-S is confirmed as a highly effective architecture for dermoscopic image analysis, achieving 98% accuracy through its compound scaling strategy and efficient feature extraction design. Its ability to simultaneously capture fine-grained textures and broader lesion structures makes it particularly well-suited for this task.

Second, ConvNeXt proves to be a strong performer at 95%, and its modernized convolutional design offers complementary strengths to EfficientNetV2. The high AUC values associated with ConvNeXt confirm its ability to produce reliable and threshold-stable class separation.

Third, while the hybrid model does not achieve the highest raw accuracy, it consistently outperforms individual models at the class level, particularly for visually similar lesion types. This finding underscores a fundamental principle in medical image classification: raw accuracy alone is not a sufficient metric. Consistent, reliable performance across all classes matters far more than peak performance on a single metric.

Fourth, the Grad-CAM analysis confirms that the model attends to clinically appropriate regions of each image, providing an important layer of validation that goes beyond numerical metrics. This interpretability is a key factor that distinguishes this system from many existing black-box classifiers and makes it a more viable candidate for real-world clinical support.

Finally, the high AUC values recorded for all four lesion classes confirm that the model's discriminative capability is genuine and robust, remaining stable across varying decision thresholds and not merely the result of favorable class distribution.

SUMMARY

This chapter presented the experimental results and analysis of all four deep learning models evaluated on the HAM10000 test set. EfficientNetV2-S achieved the highest accuracy at 98%, followed by ConvNeXt at 95%, the hybrid model at 94%, and DenseNet at 90%. AUC scores were uniformly high across all lesion classes, with bcc and akiec achieving perfect scores of 1.00. Confusion matrix analysis revealed that misclassifications were sparse and concentrated in clinically realistic patterns between melanoma and actinic keratosis. Training and validation curves confirmed stable learning without overfitting. Grad-CAM heatmaps validated that the model focuses on diagnostically relevant lesion regions, enhancing its interpretability and clinical trustworthiness. The experimental findings demonstrate that deep learning-based approaches are highly effective for automated skin lesion classification and can significantly support early disease detection. The results also highlight the importance of combining accurate prediction performance with explainable AI techniques to develop reliable and clinically meaningful computer-aided diagnostic systems.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

This mini project presented a comprehensive deep learning–based framework for multi-class skin lesion classification using dermoscopic images from the HAM10000 dataset. The study successfully addressed the challenge of automatically classifying four clinically significant skin lesion categories—melanocytic nevi, melanoma, basal cell carcinoma, and actinic keratosis—using state-of-the-art convolutional neural network architectures.

Among the three individual architectures evaluated, EfficientNetV2-S demonstrated the strongest overall accuracy at 98%, driven by its compound scaling strategy and efficient feature extraction design. ConvNeXt followed closely at 95%, offering strong spatial pattern recognition through its modernized architectural features. DenseNet achieved 90%, providing stable and consistent performance through its dense inter-layer connectivity. The proposed hybrid model, combining EfficientNetV2 and ConvNeXt through dual-branch feature concatenation, achieved 94% accuracy while delivering the most balanced and consistent performance across all four lesion categories.

A key finding of this study is that raw accuracy alone does not fully capture the value of a classification system in a medical context. The hybrid model, despite not reaching the highest individual accuracy, consistently produced more reliable class-level predictions, particularly for visually similar lesion types. This underscores the importance of multi-architecture feature fusion as a strategy for improving generalization in multi-class medical imaging tasks.

The integration of Grad-CAM visualization added an important dimension of interpretability to the proposed framework. The generated heatmaps consistently highlighted clinically relevant regions of each dermoscopic image, confirming that the model’s decisions are grounded in meaningful dermatological features rather than irrelevant background artifacts. This transparency is essential for building clinician trust in automated diagnostic systems and for enabling responsible deployment in real-world healthcare settings.

Taken together, the results of this study provide strong evidence that deep learning–based frameworks are capable of delivering both high accuracy and interpretable predictions in multi-class skin lesion classification. The proposed system has genuine potential as a computer-aided

diagnostic tool that can support dermatologists in early detection workflows and clinical decision-making.

6.2 FUTURE WORK

While the current framework demonstrates strong performance, several directions for future development could further improve its robustness, applicability, and clinical utility.

- **Transformer-Based Models:** Integrating Vision Transformers or hybrid CNN-Transformer architectures may improve feature extraction and classification accuracy.
- **Expanded Class Coverage:** Extending the framework to classify all lesion categories in the HAM10000 dataset can increase clinical applicability.
- **Class Imbalance Handling:** Techniques such as SMOTE, class-weighted loss, and GAN-based augmentation could improve minority class performance.
- **Multimodal Integration:** Combining dermoscopic images with clinical metadata may enhance diagnostic accuracy and personalization.
- **Real-Time Deployment:** Model optimization methods like pruning and quantization can support faster inference for mobile and clinical applications.
- **Larger Diverse Datasets:** Using more diverse datasets can improve generalization across different skin tones and imaging conditions.
- **Clinical Validation:** Testing the system in real clinical environments with dermatologists can verify its practical effectiveness and reliability.
- **Explainable AI Enhancement:** SHAP analysis can provide deeper insights into model decision-making and increase clinician trust in automated diagnosis systems

SUMMARY

This chapter concluded the project by summarizing the key findings and contributions. The proposed deep learning framework successfully classified four types of skin lesions from the HAM10000 dataset with strong accuracy and interpretability. EfficientNetV2-S achieved the highest individual accuracy at 98%, while the hybrid model provided the most balanced class-level performance. Grad-CAM visualizations confirmed that the model attends to clinically relevant features. Future work will focus on expanding class coverage, integrating clinical metadata, exploring transformer-based architectures, and optimizing the model for real-time clinical deployment.

APPENDIX

SAMPLE CODE

```

!pip install -q timm albumentations grad-cam shap
import os, cv2, gc, warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, label_binarize
from sklearn.metrics import *
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader, WeightedRandomSampler
import timm
import albumentations as A
from albumentations.pytorch import ToTensorV2
warnings.filterwarnings("ignore")
CSV_PATH = "/content/drive/MyDrive/kaggle dataset/HAM10000_metadata.csv"
IMG_DIRS = [
    "/content/drive/MyDrive/kaggle dataset/HAM10000_images_part_1",
    "/content/drive/MyDrive/kaggle dataset/HAM10000_images_part_2"
]
CLASS_MAP = {
    "nv": 0,
    "mel": 1,
    "bcc": 2,
    "akiec": 3
}
IDX2CLASS = {v:k for k,v in CLASS_MAP.items()}

NUM_CLASSES = 4
IMG_SIZE = 224

```

```

BATCH_SIZE = 12
EPOCHS = 30
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
CHECKPOINT = "/content/best_hybrid_model.pth"
df = pd.read_csv(CSV_PATH)
df = df[df["dx"].isin(CLASS_MAP.keys())].copy()
df["label"] = df["dx"].map(CLASS_MAP)
df["age"] = df["age"].fillna(df["age"].median())
df["sex"] = df["sex"].fillna("unknown").map({"male":0, "female":1, "unknown":2})
meta_cols = ["age", "sex"]
scaler_meta = StandardScaler()
df[meta_cols] = scaler_meta.fit_transform(df[meta_cols])
def get_path(image_id):
    for folder in IMG_DIRS:
        path = os.path.join(folder, image_id + ".jpg")
        if os.path.exists(path):
            return path
    return None
df["path"] = df["image_id"].apply(get_path)
df = df.dropna().reset_index(drop=True)
train_df, val_df = train_test_split(
    df,
    test_size=0.2,
    stratify=df["label"],
    random_state=42
)

print("Train:", len(train_df))
print("Val:", len(val_df))
train_tfms = A.Compose([
    A.Resize(IMG_SIZE, IMG_SIZE),
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.5),
    A.RandomBrightnessContrast(p=0.4),

```

```

        A.ShiftScaleRotate(p=0.4),
        A.CoarseDropout(p=0.2),
        A.Normalize(),
        ToTensorV2()
    ])
val_tfms = A.Compose([
    A.Resize(IMG_SIZE, IMG_SIZE),
    A.Normalize(),
    ToTensorV2()
])
class HAMDataset(Dataset):
    def __init__(self, dataframe, tfms):
        self.df = dataframe.reset_index(drop=True)
        self.tfms = tfms
    def __len__(self):
        return len(self.df)
    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        img = cv2.imread(row.path)
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        img = self.tfms(image=img)["image"]
        meta = torch.tensor(row[meta_cols].values.astype(np.float32))
        label = int(row.label)

        return img, meta, label
train_ds = HAMDataset(train_df, train_tfms)
val_ds = HAMDataset(val_df, val_tfms)
class_counts = train_df["label"].value_counts().sort_index().values
weights = 1.0 / class_counts
sample_weights = train_df["label"].map(lambda x: weights[x]).values
sampler = WeightedRandomSampler(sample_weights, len(sample_weights))
train_loader = DataLoader(train_ds, batch_size=BATCH_SIZE, sampler=sampler,
num_workers=2)
val_loader = DataLoader(val_ds, batch_size=BATCH_SIZE, shuffle=False, num_workers=2)

```

```

class HybridModel(nn.Module):
    def __init__(self):
        super().__init__()
        self.backbone1 = timm.create_model("tf_efficientnetv2_s", pretrained=True,
num_classes=0)
        self.backbone2 = timm.create_model("convnext_tiny", pretrained=True,
num_classes=0)
        self.meta_net = nn.Sequential(
            nn.Linear(2, 32),
            nn.ReLU(),
            nn.Dropout(0.2)
        )
        self.classifier = nn.Sequential(
            nn.Linear(1280 + 768 + 32, 512),
            nn.ReLU(),
            nn.Dropout(0.4),
            nn.Linear(512, NUM_CLASSES)
        )
    def forward(self, x, meta):
        f1 = self.backbone1(x)
        f2 = self.backbone2(x)
        fm = self.meta_net(meta)
        x = torch.cat([f1, f2, fm], dim=1)
        return self.classifier(x)

model = HybridModel().to(DEVICE)
criterion = nn.CrossEntropyLoss(label_smoothing=0.1)
optimizer = optim.AdamW(model.parameters(), lr=1e-4)
scheduler = optim.lr_scheduler.CosineAnnealingWarmRestarts(optimizer, T_0=5)
scaler = torch.cuda.amp.GradScaler()
best_acc = 0
history = {"train_acc":[], "val_acc":[]}
if os.path.exists(CHECKPOINT):
    model.load_state_dict(torch.load(CHECKPOINT))
    print("Resumed from checkpoint")

```



```

for epoch in range(EPOCHS):
    model.train()
    train_correct = 0
    for imgs, meta, labels in train_loader:
        imgs, meta, labels = imgs.to(DEVICE), meta.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        with torch.cuda.amp.autocast():
            outputs = model(imgs, meta)
            loss = criterion(outputs, labels)
        scaler.scale(loss).backward()
        scaler.step(optimizer)
        scaler.update()
        train_correct += (outputs.argmax(1)==labels).sum().item()
    train_acc = train_correct / len(train_ds)
    model.eval()
    val_correct = 0
    y_true, y_pred, y_prob = [], [], []
    with torch.no_grad():
        for imgs, meta, labels in val_loader:
            imgs, meta = imgs.to(DEVICE), meta.to(DEVICE)
            preds = []
            for flip in [None, "h", "v"]:
                x = imgs.clone()
                if flip == "h":
                    x = torch.flip(x, [3])
                elif flip == "v":
                    x = torch.flip(x, [2])
                preds.append(torch.softmax(model(x, meta), dim=1))
            probs = torch.mean(torch.stack(preds), dim=0)
            pred = probs.argmax(1)
            val_correct += (pred.cpu()==labels).sum().item()
            y_true.extend(labels.numpy())
            y_pred.extend(pred.cpu().numpy())
            y_prob.extend(probs.cpu().numpy())

```

```

val_acc = val_correct / len(val_ds)
history["train_acc"].append(train_acc)
history["val_acc"].append(val_acc)
if val_acc > best_acc:
    best_acc = val_acc
    torch.save(model.state_dict(), CHECKPOINT)
scheduler.step()
print(f"Epoch {epoch+1}/{EPOCHS} | Train Acc: {train_acc:.4f} | Val Acc:
{val_acc:.4f}")
print("Best Validation Accuracy:", best_acc)
cm = confusion_matrix(y_true, y_pred)
metrics_rows = []

for i in range(NUM_CLASSES):
    tp = cm[i,i]
    fn = cm[i,:].sum() - tp
    fp = cm[:,i].sum() - tp
    tn = cm.sum() - (tp+fp+fn)
    accuracy = (tp+tn)/cm.sum()
    sensitivity = tp/(tp+fn+1e-8)
    specificity = tn/(tn+fp+1e-8)
    metrics_rows.append([
        IDX2CLASS[i],
        accuracy,
        sensitivity,
        specificity
    ])
metrics_df = pd.DataFrame(
    metrics_rows,
    columns=["Class", "Accuracy", "Sensitivity", "Specificity"]
)
print(metrics_df)
print(classification_report(y_true, y_pred))
sns.heatmap(cm, annot=True, fmt='d')

```

```

plt.title("Confusion Matrix")
plt.show()
plt.plot(history["train_acc"], label="Train")
plt.plot(history["val_acc"], label="Validation")
plt.title("Training Stability")
plt.legend()
plt.show()
metrics_df.set_index("Class")[["Sensitivity", "Specificity"]].plot(kind="bar")
plt.title("Sensitivity vs Specificity")
plt.show()
y_bin = label_binarize(y_true, classes=[0,1,2,3])
roc_table = []
plt.figure(figsize=(8,6))
for i in range(NUM_CLASSES):
    fpr, tpr, _ = roc_curve(y_bin[:,i], np.array(y_prob)[:,i])
    roc_auc = auc(fpr, tpr)
    roc_table.append([IDX2CLASS[i], roc_auc])
    plt.plot(fpr, tpr, label=f"{IDX2CLASS[i]} AUC={roc_auc:.3f}")
plt.plot([0,1],[0,1], '--')
plt.legend()
plt.title("ROC Curves")
plt.show()
roc_df = pd.DataFrame(roc_table, columns=["Class", "ROC_AUC"])
print(roc_df)
from google.colab import files
uploaded = files.upload()
for fn in uploaded:
    img = cv2.imread(fn)
    rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    tensor = val_tfms(image=rgb)["image"].unsqueeze(0).to(DEVICE)
    meta = torch.tensor([[0,0]], dtype=torch.float32).to(DEVICE)
    with torch.no_grad():
        pred = torch.softmax(model(tensor, meta), dim=1)
        cls = pred.argmax(1).item()

```

```

        conf = pred.max().item()
    plt.imshow(rgb)
    plt.title(f"{IDX2CLASS[cls]} ({conf:.2%})")
    plt.axis("off")
    plt.show()

```

EVALAUATION METRICS

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import (
    confusion_matrix,
    classification_report,
    precision_score,
    recall_score,
    f1_score,
    roc_curve,
    auc
)
from sklearn.preprocessing import label_binarize
from math import pi
from PIL import Image
import torch
# ----- Load Best Model -----
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
# ----- Collect Predictions -----
y_true, y_pred, y_probs = [], [], []
with torch.no_grad():
    for imgs, labels in test_loader:
        imgs, labels = imgs.to(DEVICE), labels.to(DEVICE)
        outputs = model(imgs)
        probs = torch.softmax(outputs, dim=1)
        preds = torch.argmax(probs, dim=1)
        y_true.extend(labels.cpu().numpy())

```

```

    y_pred.extend(preds.cpu().numpy())
    y_probs.extend(probs.cpu().numpy())
y_true = np.array(y_true)
y_pred = np.array(y_pred)
y_probs = np.array(y_probs)
# ----- Final Test Accuracy -----
test_accuracy = np.mean(y_true == y_pred)
print("Final Test Accuracy:", test_accuracy)
# ----- Classification Report -----
print("\nClassification Report:\n")
print(classification_report(y_true, y_pred, target_names=class_names))
# ----- Precision / Recall / F1 -----
precision = precision_score(y_true, y_pred, average='weighted')
recall = recall_score(y_true, y_pred, average='weighted')
f1 = f1_score(y_true, y_pred, average='weighted')
print("Precision:", precision)
print("Recall  :", recall)
print("F1-score :", f1)
# ----- ROC Curve -----
y_true_bin = label_binarize(y_true, classes=range(NUM_CLASSES))
plt.figure(figsize=(7,6))
for i in range(NUM_CLASSES):
    fpr, tpr, _ = roc_curve(y_true_bin[:, i], y_probs[:, i])
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr, label=f"{class_names[i]} (AUC={roc_auc:.2f})")
plt.plot([0,1],[0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

# ----- Accuracy & Loss Curves -----
plt.figure(figsize=(8,4))

```

```

plt.plot(train_accs, label="Train Accuracy")
plt.plot(val_accs, label="Validation Accuracy")
plt.legend()
plt.title("Accuracy Curve")
plt.show()
plt.figure(figsize=(8,4))
plt.plot(train_losses, label="Train Loss")
plt.plot(val_losses, label="Validation Loss")
plt.legend()
plt.title("Loss Curve")
plt.show()
# ----- Radar Chart -----
precision_pc = precision_score(y_true, y_pred, average=None)
recall_pc = recall_score(y_true, y_pred, average=None)
f1_pc = f1_score(y_true, y_pred, average=None)
plt.figure(figsize=(7,7))
for i, cls in enumerate(class_names):
    values = [precision_pc[i], recall_pc[i], f1_pc[i]]
    values += values[:1]
    angles = [n/3*2*pi for n in range(3)]
    angles += angles[:1]
    plt.polar(angles, values, label=cls)
plt.thetagrids([0,120,240], ["Precision", "Recall", "F1"])
plt.legend(bbox_to_anchor=(1.3,1.1))
plt.title("Radar Chart – Per Class Performance")
plt.show()
# ----- Single Image Prediction -----
img_path = test_df.iloc[0]["image_path"]
img = Image.open(img_path).convert("RGB")
img_tensor = test_transform(img).unsqueeze(0).to(DEVICE)
with torch.no_grad():
    out = model(img_tensor)
    pred = torch.argmax(out, dim=1).item()
print("Single Image Prediction:", class_names[pred])

```

CONFUSION MATRIX

```

from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import numpy as np

cm = confusion_matrix(all_true, final_labels)
plt.figure(figsize=(6,5))
plt.imshow(cm, cmap='Blues')
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.xticks(range(NUM_CLASSES))
plt.yticks(range(NUM_CLASSES))
for i in range(NUM_CLASSES):
    for j in range(NUM_CLASSES):
        plt.text(j, i, cm[i,j], ha="center", va="center")
plt.show()

from sklearn.metrics import precision_score, recall_score, f1_score

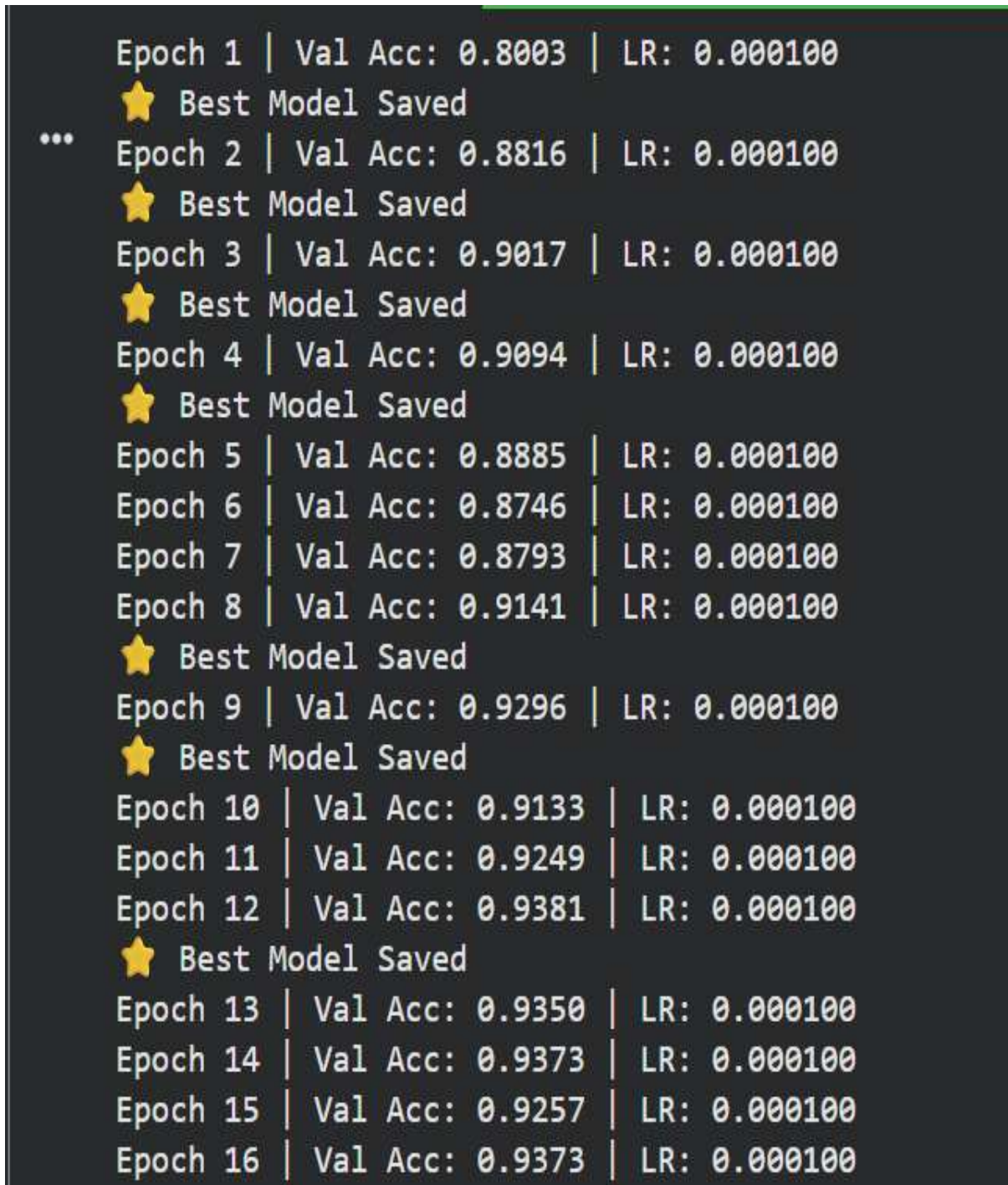
precision = precision_score(all_true, final_labels, average="weighted")
recall = recall_score(all_true, final_labels, average="weighted")
f1 = f1_score(all_true, final_labels, average="weighted")
accuracy = acc

metrics = [precision, recall, f1, accuracy]
labels = ["Precision", "Recall", "F1-Score", "Accuracy"]
angles = np.linspace(0, 2*np.pi, len(labels), endpoint=False)
metrics = np.concatenate((metrics,[metrics[0]]))
angles = np.concatenate((angles,[angles[0]]))
plt.figure(figsize=(6,6))
ax = plt.subplot(111, polar=True)
ax.plot(angles, metrics, 'o-')
ax.fill(angles, metrics, alpha=0.25)
ax.set_thetagrids(angles[:-1]*180/np.pi, labels)

```

APPENDIX -2

SCREENSHORTS



Epoch 1	Val Acc: 0.8003	LR: 0.000100
★ Best Model Saved		
...		
Epoch 2	Val Acc: 0.8816	LR: 0.000100
★ Best Model Saved		
Epoch 3	Val Acc: 0.9017	LR: 0.000100
★ Best Model Saved		
Epoch 4	Val Acc: 0.9094	LR: 0.000100
★ Best Model Saved		
Epoch 5	Val Acc: 0.8885	LR: 0.000100
Epoch 6	Val Acc: 0.8746	LR: 0.000100
Epoch 7	Val Acc: 0.8793	LR: 0.000100
Epoch 8	Val Acc: 0.9141	LR: 0.000100
★ Best Model Saved		
Epoch 9	Val Acc: 0.9296	LR: 0.000100
★ Best Model Saved		
Epoch 10	Val Acc: 0.9133	LR: 0.000100
Epoch 11	Val Acc: 0.9249	LR: 0.000100
Epoch 12	Val Acc: 0.9381	LR: 0.000100
★ Best Model Saved		
Epoch 13	Val Acc: 0.9350	LR: 0.000100
Epoch 14	Val Acc: 0.9373	LR: 0.000100
Epoch 15	Val Acc: 0.9257	LR: 0.000100
Epoch 16	Val Acc: 0.9373	LR: 0.000100

Figure A.1 Training the Model

Figure A.1 represents the validation performance of the hybrid deep learning model, showing high accuracy and excellent classification results across all classes.

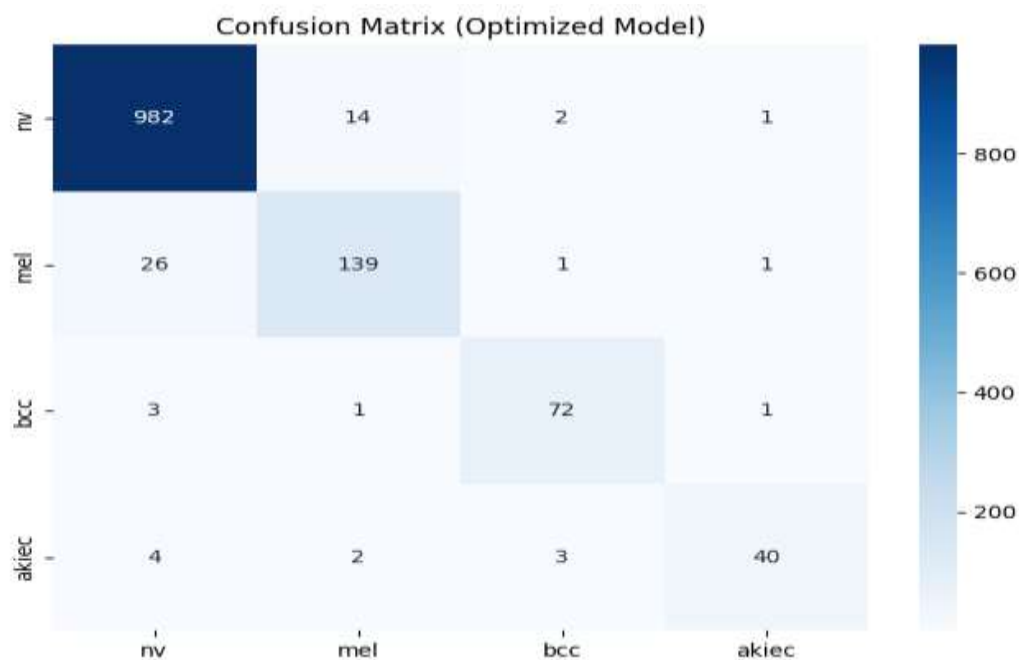


Figure A.2 Confusion Matrix

Figure A.2 represents the confusion matrix of the optimized hybrid deep learning model, illustrating the classification performance across the four skin lesion classes: *nv*, *mel*, *bcc*, *akiec*.

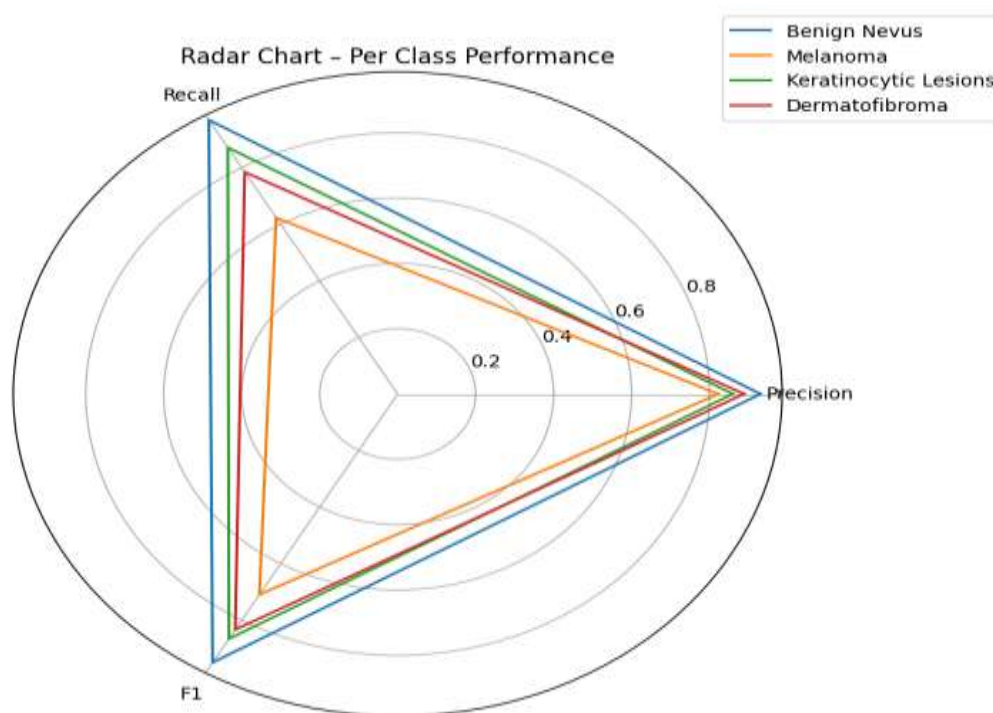


Figure A.3 Radar Chart

Figure A.3 represents the radar chart analysis of per-class performance for the skin lesion classification model.

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SUSTAINABLE DEVELOPMENT GOALS

Goal 3: Good Health and Well-Being

The proposed project primarily aligns with SDG 3, which focuses on ensuring healthy lives and promoting well-being for people of all ages. The Deep Learning Based Multi-Class Skin Lesion Diagnosis System contributes to this goal by enabling early detection of skin diseases, improving diagnostic accuracy, and supporting healthcare professionals in effective decision-making.

The project contributes to SDG 3 in the following ways:

1 .Early Detection of Skin Diseases

The system analyzes dermoscopic images to identify and classify various types of skin lesions, including melanoma and other skin conditions. Early detection of skin cancer significantly increases the chances of successful treatment and reduces mortality rates. By identifying diseases at an initial stage, the system helps in timely medical intervention and improved patient outcomes.

2. Use of Advanced Deep Learning Techniques in Healthcare

The project utilizes modern deep learning architectures such as Convolutional Neural Networks (CNN), EfficientNet, DenseNet, and ConvNeXt for accurate image classification. These techniques enhance the system's ability to extract meaningful features from medical images, thereby improving the reliability and efficiency of disease diagnosis in dermatology.

3. Support for Clinical Decision-Making

The proposed system assists dermatologists and medical practitioners by providing automated classification results. It reduces the time required for manual diagnosis and minimizes the chances of human error. This support enables healthcare professionals to make faster, more informed, and accurate clinical decisions.

4. Improved Accessibility to Healthcare Services

The model can be deployed in hospitals, diagnostic centers, and remote healthcare systems. It enables large-scale screening of skin diseases, especially in areas where access to specialized dermatologists is limited. This contributes to equitable healthcare delivery and promotes early diagnosis in underserved regions.

5. Scalable and Efficient Healthcare Solution

The system is designed to handle large datasets and can be integrated into digital healthcare platforms for continuous monitoring and diagnosis. Its scalability makes it suitable for real-time applications, helping healthcare systems manage increasing patient loads effectively.

Overall, this project supports SDG 3: Good Health and Well-Being by enhancing early detection of skin diseases, improving diagnostic accuracy, supporting medical professionals, and promoting accessible healthcare solutions. It contributes to better health outcomes and strengthens the role of technology in modern healthcare systems.

SIGNATURE OF THE SUPERVISOR